



The University of Zambia

In association with

Zambia Centre for Accountancy Studies

BACHELOR OF ARTS FINANCIAL SERVICES

**BFS421 RESEARCH METHODS
MODULE**

Year 4

Semester 1

Author: Dr Kelvin Mukolo Kayombo



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1.0 INTRODUCTION



Welcome to the BFS421 Research Methods course. I am delighted that you are now in a position to develop a piece of independent research, an exciting new challenge for any undergraduate student. This module will introduce you to the research skills, knowledge and understanding that will enable you to prepare your dissertation proposal and ultimately complete your dissertation. The module has been carefully designed to support your studies and to guide you in the research process. It is a step-by-step guide to the key elements of a research dissertation that hopefully provides you with a framework within which to develop your own ideas and interests.

Research can be a lonely and sometimes frustrating activity; but it can also be a source of great personal satisfaction. You have in this module the opportunity to develop your ideas about a topic that interests you.

You set the boundaries and objectives; you decide what to read and how to collect data. You own this project!

Having said this, there is support and guidance. The Moodle platform provides you with an opportunity to share your ideas (and frustrations!). It hopefully gets you thinking about your research aims and your research strategy. After your proposal has been accepted, you will have a dissertation supervisor who will provide support and feedback as you embark on the research itself.

Completing the reading and designated activities are essential to enable you to complete the proposal and the dissertation successfully.

In the remainder of this introductory chapter you are given advice and guidance on the following:

- The course aims and objectives
- The structure of the course
- Assessment framework
- Prescribed and recommended textbooks
- Study skills
- How much time you need to spend on this course
- Who to contact if you need any help

It is important to understand all of these at the beginning to ensure that you are able to get the most out of the course.

1.1 MODULE AIM



The aim of this module is to enable you understand and use quantitative and qualitative research techniques to answer a research question, as appropriate to the final-year undergraduate level. As you are aware, you will be required to undertake a research project or dissertation in the second semester of your final year on the degree programme. This requires you to understand the main components of a dissertation such as an abstract, introduction, literature review, methodology, findings and analysis, and conclusions. This module will introduce you to these components and also challenge you to start thinking about your research next semester.

1.2 OBJECTIVES



By the end of this module, you should be able to:

1. Understand the theoretical underpinning and practical implications of research conducted at final-year undergraduate level.
2. Utilise qualitative and quantitative research methods to answer a research question.
3. Compare alternative research approaches and evaluate their uses and limitations.
4. Understand the key concepts and tasks involved in a research project and how these interrelate in a complementary way to form a coherent research proposal.
5. Process and analyse research data using data analysis software.

1.3 ASSESSMENT DETAILS



1.3.1 Assessment structure

In order to ensure that the learning outcomes have been achieved, you are assessed as follows:

Continuous assessment	50%
2 assignments of equal weight	
Research Proposal	<u>50%</u>
Total	<u>100%</u>

1.3.2 Assessment description and deadlines

Assessment Title	Weight towards final grade	Word count	Due Date	Return Date
Assignment 1: Literature Review and objectives/hypothesis & research questions	25%	1,000 words (+/-10%)	Second week of April or October. Exact date to be confirmed (TBC).	Last week of April or October
Assignment 2: Research Methodology	25%	1,000 words (+/-10%)	Second week of May or November. Exact date TBC.	Last week of May or November
Research Proposal	50%	2,000 words (+/-10%)	Third week of June or December. Exact date TBC.	Official date of release of results

- 1. Assignment 1 is a critical appraisal of a literature review section. This assignment is issued at commencement of the course.*
- 2. Assignment 2 is a critical appraisal of a research methodology section. This assignment is issued during the first week of April or October*
- 3. The Research Proposal should have 2,000 words +/- 10%, Times New Roman font size 12, 1.5 spacing with justified text.*

1.3.2 Assessment submission through TurnItIn

As you know by now all course-work on your degree programme is submitted through TurnItIn; and this course is no exception. Your course-work is therefore electronically scanned for plagiarism. Note also that only paid-up students are registered on TurnItIn, hence you may be unable to submit your course-work if you have not paid your tuition fees.

1.4 READINGS

1.4.1 Prescribed Reading

You need to read at least one of the following text books on business research to successfully complete this course and the dissertation thereafter.



1. Bryman, A. and Bell, E. (2011) *Business Research Methods*, 3rd Edition. New York: Oxford University Press.
2. Saunders, N. K. M, Thornhill, A. and Lewis, P. (2009) *Research Methods for Business Students*, 5th Edition, London: Prentice Hall.

1.4.2 Recommended Reading



You may refer to the following text books on business research as part of your studies for this course and also as you undertake the research next semester.

1. Allison, B. (2004) *The student's guide to preparing dissertations and theses*. Routledge.
2. Bell, J. (2008) *Doing your research project: a guide for first time researchers in education and social science*. Open University Press.
3. Collins, J. & Hussey, R. (2003) *Business Research: A practical guide for undergraduates and Postgraduates students*. Palgrave
4. Fisher, C. (2004) *Researching and writing a dissertation for business students*. FT Prentice Hall
5. Gash, S. (2000) *Effective literature searching for research*. Gower
6. Hart, C. (2001) *Doing a literature Search*. Sage.
7. Kumar, R. (1999) *Research methodology: a step-by-step guide for beginner*. Sage.
8. Patton, R. (2001) *Qualitative Evaluation & Research Methods*. Sage.

1.5 TIME FRAME



You should spend a minimum of 120 hours on this module. This time should be spent on reading the contents of this unit, reading the prescribed reading and on attempting the activities in the module, as well as the self-check and discussion questions in the prescribed readings, itemized in each unit.

1.6 STUDY SKILLS



As a distance education student, you will meet a lot of challenges in your studies, particularly that you will not always have a lecturer and fellow students to consult. You need, therefore, to develop a strategy for yourself which will make you succeed. Firstly, draw a working time-table and stick to it. In doing this, remember the power of ‘one hour per day’. If you commit yourself and persistently work on something, anything, for just one hour every day, you will become an expert at it.

Secondly, work on all the tasks and assignments. As a minimum, ensure to undertake the required reading specified in this module. Remember that reading this module alone is not adequate preparation for your studies. Attempting the self-check and discussion questions is important as it helps you to consolidate what you have read. Self-discipline is very important since no one is monitoring what you are doing on a daily basis. Submit all the work that you need to submit on time.

Finally, you should attend the residential school at the end of the semester. This is an opportunity for you to meet your course tutor in person and clarify any issues you have. The residential school also provides you with an opportunity to network with other students.

1.7 NEED HELP?



If you need help on the module, please use the following contacts:

Course Tutor: Dr Kelvin Mukolo Kayombo

Email: kkayombo@zcas.edu.zm; kelvin.kayombo@gmail.com

Zambia Centre for Accountancy Studies (ZCAS)

Dedan Kimathi Road, P O Box 35243, Lusaka, Zambia

Tel: +260 1 232093/5, Fax: +260 1 222542

2.0 UNIT ONE: ROLE OF RESEARCH AND ETHICS

2.1 INTRODUCTION



This unit has two main themes namely the nature and purpose of research, particularly business and management research, and the importance of ethics in research. Research as defined and used in this course is different from the everyday use of the term in which ‘research’ refers to collection of data or information without a definitive purpose or without thorough interpretation or perhaps just for marketing purposes. This is not research in the true sense of the word. To be considered as research, an activity should have a number of features such as systematic collection of data, systematic interpretation of data, and a clear objective of finding out things. In the context of business and management, research should not only result in more knowledge and understanding, but also help in solving organizational problems.

Research ethics is concerned with adopting behaviours that do not harm research participants as well as users of the research findings. The question of moral and responsible behavior is pervasive to the research process and needs to be considered throughout your research as there is potential to cause harm to your research subjects throughout your research project i.e. from conception of the research idea to writing up the dissertation.

2.2 AIM



The aims of this unit are to help you appreciate the meaning and purpose of business and management research, and the importance of good ethical behaviour in research.

2.3 OBJECTIVES



By the end of this unit, and having completed the prescribed readings and activities, you should be able to:

1. Explain the nature and purpose of research, including business research.
2. Describe the different stages of the research process.

3. Identify a potential topic for your research.
4. Explain the role of ethics in business research e.g. confidentiality, plagiarism, misrepresentation of facts, and cheating.

2.4 TIME REQUIRED



You should spend a minimum of 5 hours on this unit. This time should be spent on reading the contents of this unit, the prescribed reading, and on attempting the activities in the text book as indicated below.

2.5 REFLECTION



Research is a process that we all go through; we are constantly collecting and interpreting data, both in our work life and in our home life. Think about a research activity in which you have been involved. This might be a formal investigation of a problem or opportunity at work or it may be research associated with a personal investment such as the purchase of a house or a car. Now answer the following questions.

- What was the problem/issue that you were investigating?
- How did you gather information/data relating to the problem/issue?
- Where did you get your data from?
- How did you analyse your data? For example, did you use statistics or numbers to help you make sense of your information?
- Was your research successful? Explain the reasons for its success or failure.
- Did you encounter any problems? List and explore the problems. If you did not experience any problems, think about the reasons why your research was problem-free!
- What have you learned from your previous research experience? For example, what are you good at and what did you find difficult?
- How might you improve any future research that you undertake?

- What action will you take to help you submit a successful research proposal and dissertation on your undergraduate programme?

2.6 NATURE AND PURPOSE OF RESEARCH

This section addresses two issues, namely, the nature and purpose of any research activity, and the nature and purpose of business and management research.

2.6.1 The nature of research

Saunders, Lewis and Thornhill (2009, p.5) define research as ‘something that people undertake in order to find out things in a systematic way, thereby increasing their knowledge’. The two key terms from this definition are ‘systematic way’ which implies logical relationships as a basis for research and ‘to find out things’ which suggests that research must have a specific purpose. To be considered as research, an activity should therefore have a number of features such as systematic collection of data, systematic interpretation of data, and a clear objective of finding out things. The common use of the term ‘research’ when referring to collection of data or information without a definitive purpose or without thorough interpretation or perhaps just for marketing purposes is not research in the true sense of the word (Walliman, 2005).

2.6.2 The nature of business and management research

Business and management research is defined as ‘undertaking systematic research to find out things about business and management’ (Saunders *et al.*, 2009, p.5). This type of research can be distinguished from other forms of research (e.g. research in science, medicine and engineering) because it tends to be trans-disciplinary, is normally only carried out if there are personal or commercial benefits, and also if the findings can have practical implications. They further argue that business and management research should not only result in more knowledge and understanding, but also help in solving organizational problems. However, the extent to which research can be classified as business and management research depends on its purpose and context and can be mapped on a continuum (p.8). Such a continuum would have basic or fundamental or pure business and management research at one end (i.e. research carried out purely for academic purposes e.g. to understand business and management processes and outcomes without practical application). Research that has direct and immediate relevance and impact on business and management (i.e. not for academic purposes) would be placed at the other end of the continuum.

2.6.3 REQUIRED READING 2.1



Now read Saunders *et al.* (2009, pp.2 – 9).

2.7 RESEARCH PROCESS

The process of carrying out research is normally represented as a series of stages that you need to go through to complete your research project, with the following as the order of the main or common stages involved (Saunders *et al.*, 2009, p.10):

Formulate and clarify your research topic



Critically review the literature



Design the research



Collect data



Analyse data



Write up the dissertation

However, the research itself is iterative in nature and you will need to revisit all the stages several times to revise, clarify, and polish up issues.



2.7.1 Required reading 2.2

Now read Saunders *et al.* (2009, pp.10 – 11).

ACTIVITY



Now attempt the self-check and discussion questions on page 15 in Saunders *et al.* (2009, p.15).

2.8 IDENTIFYING A RESEARCH TOPIC

The issues involved in generating and refining research ideas to come up with a research topic and research design will be dealt with in great detail in the next unit. Nevertheless, as Bryman and Bell (2011, p.73) observe, it is worthwhile to start thinking about your research topic early and to give yourself a great deal of time for this activity. There could be something about the other courses you are taking this semester or in your earlier courses that is of interest to you; and the right time to think about what it is you really want to investigate is while you still have fresh memories of the subject area.

Although the idea of choosing your own research topic appears exciting to you on face value, making the decision, refining it and coming up with a researchable topic may be one of the greatest challenges you will face in your studies. You therefore need to start thinking about possible issues you want to investigate. These could be subject areas of interest from your studies, organizational challenges that your employer is experiencing, a business opportunity you wish to pursue in the future, or a problem being faced in your community. Having preliminary discussions about any potential research area with relevant parties, even at this early stage, can help you to clarify and focus your thinking.

2.9 ACTIVITY 2.1



Now write down three ideas that you have for your research in the spaces below.

- 1.....
- 2.....
- 3.....

2.10 ETHICS IN BUSINESS RESEARCH

Saunders *et al.* (2009, pp.183 – 184) state that ‘In the context of research, ethics refers to the appropriateness of your behavior in relation to the rights of those who become the subject of your work, or are affected by it.’ Bryman and Bell (2011, p.122) have a similar view as they state that ethical issues in business and management research are concerned

with how you should treat the people on whom you conduct your research and whether there are activities that you should or should not undertake in relation to those people. The question of moral and responsible behavior is pervasive to the research process and needs to be considered throughout your research i.e. from conception of the research idea to writing up the dissertation, as there is potential to cause harm to your research subjects.

The general ethical issues that are likely to arise during your research project are outlined by Saunders *et al.* (2009, pp.185 – 186) as:

- Privacy of possible and actual participants;
- Voluntary nature of participation and the right to withdraw partially or completely from the process;
- Consent and possible deception of participants;
- Maintenance of the confidentiality of data provided by individuals or identifiable participants and their anonymity;
- Reactions of participants to the way in which you seek to collect data, including embarrassment, stress, discomfort, pain and harm;
- Effects on participants of the way in which you use, analyse and report your data, in particular the avoidance of embarrassment, stress, discomfort, pain and harm;
- Behavior and objectivity of you as researcher.

2.10.1 Required reading 2.3



Now read Saunders *et al.* (2009, pp.183 – 201).

2.11 ACTIVITY 2.2



Now attempt the self-check and discussion questions in Saunders *et al.* (2009, pp.202 – 203).

2.12 SUMMARY



This unit has explored the meaning of research as applied in business and management (where research should ideally result in an increase in the body

of knowledge in a particular field as well as have practical benefits), and distinguished the use of the term as applied in everyday life (where the term ‘research’ is largely used to gain publicity by referring to bogus surveys). You have also been introduced to the stages that are involved in undertaking your research project, together with a brief outline of the issues that you should consider in identifying your research topic. Finally, the issues that you should consider to ensure that your behaviour does not harm your research subjects or users have been discussed.

3.0 UNIT TWO: RESEARCH PROPOSAL

3.1 INTRODUCTION

At the end of this course, you will be required to submit a research proposal that will form the foundation for your dissertation next semester. It is therefore important that you understand the contents of a research proposal.



The research proposal is the road map for your research. It should clearly demonstrate your capacity and ability to undertake the research. Your research topic, research objectives and research questions are key components of your proposal and this unit will take you through the stages of generating and refining them. These components of the research proposal determine everything else about your project, including your research strategy, data collection, and data analysis techniques.



3.2 AIM

The aims of this unit are to help you understand the contents of a research proposal and appreciate the issues involved in generating a research topic, research objectives, and research questions.

3.3 OBJECTIVES



By the end of this unit, and having completed the prescribed readings and activities, you should be able to:

1. Explain the content of a research proposal e.g. topic, background/context, research question/aim/objectives, literature review, methodology, role of ethics, timescale and resources.
2. Generate and refine research ideas.

3. Generate a research topic and evaluate the attributes of a good research topic.
4. Design and evaluate research aims, objectives and questions

3.4 TIME REQUIRED



You should spend a minimum of 20 hours on this unit. This time should be spent on reading the contents of this unit, reading the prescribed reading [Chapter 2 in Saunders *et al.* (2009) and/or Chapter 3 in Bryman and Bell (2011)] and on attempting Activity 3.1 as well as the self-check and discussion questions in the prescribed readings itemized above.

3.5 REFLECTION



Choosing a topic can often delay you taking action on developing your research proposal and ultimately delay your completion of the programme. Consider the following in trying to come up with a topic for your research, and write down your answers in the spaces provided.

- a) Is there any issue which is of particular interest to you that you have come across during your studies?

.....

- b) Can your employer suggest a topic for you? If you are in employment, your employer may have a problem that they want you to investigate. It is worth keeping in mind that you must meet the academic criteria for the dissertation, which may be different from the outcomes an employer may expect or those of a business proposal.

.....

- c) Can you research an idea for a business opportunity? If you contemplate running a business in future, carrying out research in the area of business will help you clarify many issues and identify associated risks.

.....
.....
.....

- d) Are there any socio-economic issues in your community that you wish to investigate? Your community could be facing problems that your research could help resolve.

.....
.....
.....

3.6 CONTENTS OF A RESEARCH PROPOSAL

As you are aware, you will be required to submit a research proposal as part of the assessment for this course. The research proposal must be of a reasonable standard for you to proceed to undertake the research project next semester.

Bryman and Bell (2011, p.84) posit that a research proposal is a road map for your research project. A research proposal encourages you to think and clarify issues that will affect the execution of your project. Saunders *et al.* (2009, pp.41 – 42) outline the purposes of the research proposal as follows.

- ✓ Writing the proposal helps you to clarify your thoughts and organize your ideas into a coherent statement of research intent.
- ✓ The research proposal can be used as a basis for discussion with your supervisor and for you to obtain feedback on the viability of your proposed study.
- ✓ Once approved, the research proposal becomes the authority for you to undertake the research i.e. a contract between you and your employer or supervisor.

The standard contents of a research proposal are outlined by Saunders *et al.* (2009, pp.42 – 45) and Bryman and Bell (2011, p.85) as follows.

- ✓ Research topic – gives a general overview of what your research is all about.

- ✓ Background or literature review – this is a justification for your research in the context of the literature in the field of your research.
- ✓ Research aims, questions, and objectives – these further clarify what your study is about by outlining specific issues you will investigate and the precise outcomes of the research project.
- ✓ Research methods – you need to outline and justify how you will achieve your research objectives and answer your research questions. This part of your research proposal includes justification of your chosen research paradigm, research strategy and data collection, and analysis techniques.
- ✓ Timescale – you need to carry out your research within a given timeframe. To achieve this, it is important that you outline the main activities that you will undertake, together with the deadlines for implementing them. A Gantt chart is a useful tool for scheduling these activities.
- ✓ Resources – you need to demonstrate that you have the necessary financial resources, equipment, access to data and skills to successfully undertake the research.
- ✓ Ethical issues – in this section, you need to demonstrate that you have considered how your research activities could harm your research participants and users of your research outcomes, including steps you will take to mitigate the potential harm.
- ✓ References – this is a list of all literature and information sources that you will have used and referred to in support of your proposal.

3.7 GENERATING AND REFINING RESEARCH IDEAS

Although the idea of choosing your own research topic appears exciting to you on face value, making the decision, refining it and coming up with a researchable topic may be one of the greatest challenges you will face in your studies. You therefore need to start thinking about possible issues you want to investigate.

3.7.1 Sources of research ideas

Potential sources for research ideas include the following.

- a) Anticipated career path – if you wish to pursue a banking career, for example, then undertaking research that involves a financial institution could broaden your knowledge and understanding of this sector. You could then evaluate the financial

and business performance of several banks or investigate the impact of ICT on bank performance.

- b) Business opportunity you are pursuing or wish to pursue in the future – if you are running a business or intend to do so in future, then undertaking research in that area of business may help you to run the business better. For example, investigating the challenges faced by small and medium-sized enterprises (SMEs) in Zambia may help you understand sources of SME funding, available government support to SMEs, common causes of SME failure etc. (among others). This knowledge can be useful in setting up and running your own SME.
- c) Organisational challenges that your employer is experiencing – your employer may suggest a research idea to you or you may wish to investigate some aspect of your organisation, such as a problem in a particular department or some aspect of your job.
- d) Previous dissertation topics – reviewing topics researched by other students or academics can help you identify potential areas for your own research (Saunders *et al.*, 2009, p.25). For example, if previous research investigated the impact of total quality management (TQM) on employee motivation, then you could simply change the context by investigating the impact of TQM on organizational performance. A number of topics researched by other students on different undergraduate programmes are listed in Appendix 2 at the end of this module.
- e) Problems being faced in your community – your community could be facing a problem that research can help to resolve.
- f) Scanning the media – current issues that can be researched further can be identified by reading print and online newspapers (Saunders *et al.*, 2009, p.27).
- a) Subject areas of interest from your studies – you could investigate theories, models or concepts that you found fascinating or that you did not agree with during your studies. For example, you may not agree that performance appraisal enhances organizational performance or that the digital divide really exists. You could then investigate these concepts by researching them in a real context or reading up some previous research works to see gaps in them which you can research.

3.7.2 Refining research ideas

You need to turn your research idea into a researchable topic. The following techniques can help you to operationalize your research idea.

- b) Discussing the research idea with your research supervisor or lecturer can help you to clarify and focus your thinking.
- c) Using the Delphi technique in which you conduct a group discussion with parties that are interested or involved in your research to narrow down your research idea (Saunders *et al.*, 2009, pp.29 – 30).
- d) Undertaking a preliminary study e.g. the initial literature review (Saunders *et al.*, 2009, p.30) can help in contextualizing your research.

3.8 ATTRIBUTES OF A GOOD RESEARCH TOPIC

As you turn your research idea into a researchable topic, note the following attributes of a good research topic as posited by Saunders *et al.* (2009, pp.22 – 23).

- a) Your research must meet the university's requirements and assessment criteria. In your case, your topic should be in a business-related area such as finance, accounting, business administration, human resource management, marketing, entrepreneurship, among others etc. The word count should be 10, 000 +/- 10%. The recommended font is 12 point, Times New Roman with line spacing of 1.5, and your in-text citations and referencing should be based on the Harvard Referencing System.
- b) Your topic must be congruent with your personal interests, capabilities, experience and limitations.
- c) You need to have or be able to acquire the skills necessary to research the topic.
- d) A good topic is one that can be researched using the resources (such as time and finances) that you are able to acquire within the research period.
- e) Access to data is a major issue in any research project. Hence you cannot choose a topic to do with the motivation of staff in the Zambia Air Force or United States embassy if you do not work there, and it is therefore highly unlikely that you will be given access to sensitive data about employee morale.
- f) A good research topic should be based on the literature, and the research should be capable of providing fresh insights into the topic i.e. your research should contribute to the existing body of knowledge on the topic (originality or further contribution to knowledge).

Even though a topic title can be expressed as a question, it is better to avoid this because such a topic title could be confused with the research question(s). Instead a topic title should be expressed as a statement which clearly indicates that you want to find out something. Please see Appendix 2 for examples of topic titles.

3.8.1 Activity 3.1



Now go back to the three research ideas/topics that you outlined in ACTIVITY 2.1 above and ask yourself the following questions and write your answers in the spaces provided.

- ✓ Is the topic in a business-related area such as finance, accounting, business administration, human resource management, marketing, entrepreneurship etc.?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....
- ✓ Is the topic in line with your personal interests, capabilities, experience and limitations?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....
- ✓ Do you have or can you develop the necessary skills e.g. ICT within the research period to research the topic?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....
- ✓ Do you have or can you acquire the necessary resources (such as time and finances) within the research period?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....
- ✓ Will you be able to access the data or information that you need for this research?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....
- ✓ Is there adequate literature e.g. text books and journal articles on this topic?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....

- ✓ Is the research capable of providing fresh insights into the topic?
 - Topic 1.....
 - Topic 2.....
 - Topic 3.....

Now evaluate the answers you have given to the questions on each topic and determine the topic that seems most suitable for your research project. Write the topic title here:.....

3.9 RESEARCH OBJECTIVES AND QUESTIONS

Once you decide on your research topic, you then need to break it down into manageable components such as research questions and/or research objectives. Carrying out research is really more or less like eating a cow; once you slaughter a cow, you need to cut it into smaller manageable pieces (firstly into the major limbs to facilitate handling and transportation, and then into even smaller pieces that can fit in a pot, deep freezer, microwave etc.). In the context of your research, this breakdown structure implies that you would need to break down your research topic into research objectives (ROs), and those research objectives further broken down into research questions (RQs).

ROs derive from the research topic and indicate, in broad terms, what your research is intended to achieve. To derive the ROs, read your research topic several times and ask yourself what it is you want to establish or find out in your research. What are the likely major outcomes of your research? What is your study likely to uncover? What will your research project contribute to the existing body of knowledge in your topic area? Please note that two or three ROs are likely to be sufficient for your research.

RQs derive from ROs and are the foundation for the research. RQs operationalise or break down ROs into specific activities that can be implemented. Each RO should be broken down into several RQs. To derive your RQs, read your RO several times and think about the questions you can ask yourself in order to meet that RO. What do you need to do or know for you to achieve your RO? Two to four RQs should be sufficient to help you establish your ROs. Once you design your RQs, then everything you do i.e. from the literature you review, the business models you choose to use, the data you collect and how you collect and analyse it etc. will be determined by those RQs. Please note that RQs are

actual questions that should have a question mark at the end and should start with or utilize words such as why, what, how, who and when.

For example, if your topic title is: *An evaluation of the business and financial performance of Zambia National Commercial Bank Plc. over the last five years*, then your ROs would be:

RO1: To establish the financial performance of Zambia National Commercial Bank Plc. in terms of capital adequacy, asset quality, management capability, earnings and liquidity over the last five years.

RO2: To establish the business performance of Zambia National Commercial Bank Plc. in terms of growth over the last five years.

The RQs necessary to help you establish these ROs would then be:

RQ1: How adequate has the capital of Zambia National Commercial Bank Plc. been over the last five years? This research question addresses research objective RO1.

RQ2: What has been the asset quality of Zambia National Commercial Bank Plc. over the last five years? This research question addresses research objective RO1.

RQ3: How effective has the management of Zambia National Commercial Bank Plc. been in running the business over the last five years? This research question addresses research objective RO1.

RQ4: To what extent has the Zambia National Commercial Bank Plc. achieved its earnings targets over the last five years? This research question addresses research objective RO1.

RQ5: How well has the Zambia National Commercial Bank Plc. managed its cash-flows over the last five years? This research question addresses research objective RO1

RQ6: How did the Zambia National Commercial Bank Plc. grow its market share in the Zambian banking sector over the last five years? This research question addresses research objective RO2.

RQ7: What growth did the Zambia National Commercial Bank Plc. achieve in terms of its product portfolio over the last five years? This research question addresses research objective RO2.

Note that different universities (and text books on research methods) have differing breakdown structures. While some use the Topic $\implies$ ROs $\implies$ RQs approach, others prefer the Topic $\implies$ RQs $\implies$ ROs breakdown structure.

3.9.1 Activity 3.2



In respect of the topic you chose in Activity 3.1 above, design two or three research objectives and write them down here. Think about the questions that you need to ask yourself regarding each RO and write them down against each objective.

RO1.....
.....

RQ1.....

RQ2.....

RQ3.....

RO2.....
.....

RQ4.....

RQ5.....

RO3.....
.....

RQ6.....

RQ7.....

This activity is very important and I strongly advise you should make an effort to discuss your research topic, research objectives and research questions with your lecturer and get

his/her feedback. The reason is that once you agree (or disagree!) on these fundamental aspects of your research, you cannot change them during the research next semester.

3.10 REQUIRED READING 3.1



Now read Chapter 2 in Saunders *et al.* (2009) and/or Chapter 3 in Bryman and Bell (2011).

3.11 SUMMARY



This unit has introduced you to the issues that are involved in writing a research proposal. The contents of a research proposal have been described and include the research topic, background/context/literature review, research question/aim/objectives, literature review, research methodology, role of ethics, timescale, and resources.

The factors that you need to consider in generating ideas for your research topic such as your anticipated career path, potential business opportunities, problems faced by your organisation/community, previous dissertation topics, topical issues, and your own personal interest have been highlighted. You have also been introduced to the attributes of a good research topic. These characteristics are important in helping you refine your research idea into a researchable topic, and they include meeting the university's specifications, consideration of your own personal interests, skills and resources, access to data, and availability of a theoretical framework upon which to base your research.

Furthermore, this unit has introduced you to the need to breakdown your research topic into research objectives (outcomes of your research) and then further breakdown those research objectives into research questions (specific queries you need to answer to establish your objectives). Such a breakdown structure helps to make your research more manageable.

The rest of the key components of a research proposal (and the dissertation itself) such as the literature review (Unit 3), research methodology (Unit 4) and findings and analysis (Unit 5) are discussed in detail in the sections below.

4.0 UNIT THREE: LITERATURE REVIEW

4.1 INTRODUCTION



The literature review is one of the major stages in your research project. It will certainly consume a significant amount of your research project time, and it involves a lot of writing. The literature review should be based on your research objectives and questions, and it should feed into your research methodology. In order to demonstrate awareness and understanding of the body of knowledge in your research area, you need to be critical in your review of that literature and not accept everything you read on face value. As you present your literature review, you need to be mindful of the dangers of plagiarism. The advice that I give you in this unit is designed to assist you produce a critical literature review chapter in a manner that avoids plagiarism.

4.2 AIM



This unit aims to introduce you to the issues involved in carrying out a critical review of the literature in your area of study, and how to process and present that literature review without plagiarizing the work of others.

4.3 OBJECTIVES



By the end of this unit, and having completed the prescribed readings and activities, you should be able to:

1. Explain the purpose of a literature review.
2. Identify literature sources.
3. Carry out a literature search including defining parameters, generating key words/terms, and use of databases in online libraries.
4. Evaluate the literature and assess relevance and sufficiency.

5. Critically review the literature in terms of theoretical underpinnings, expert opinions, and your own evaluation.
6. Explain in-text citations, bibliography and referencing, and demonstrate the use of the Harvard and APA systems of referencing.

4.4 TIME REQUIRED



You should spend a minimum of 20 hours on this unit. This time should be spent on reading the contents of this unit, reading the prescribed reading [Chapter 3 in Saunders *et al.* (2009) and/or Chapter 4 in Bryman and Bell (2011)] and on attempting Activity 4.1 – 4.3 as well as the self-check and discussion questions in the prescribed readings itemized above.

4.5 REFLECTION



I'm sure that you are aware of the term 'reinventing the wheel' and the need for you not to reinvent the wheel in many circumstances. Suppose that you want to research the competitiveness of a company in a given industry, what business models would you use? How would you relate the use of such models to the concept of 'reinventing the wheel'? Now think about your research project (and the literature review in particular) and how you should not reinvent the wheel. How can critically reviewing the literature on your topic help you not to reinvent the wheel?

4.6 PURPOSES OF A LITERATURE REVIEW

4.6.1 Difference between literature and data/information

You need to be clear about the distinction between the terms 'data'/'information' and 'literature' as used in research because both are necessary for your research project and yet they mean different things. In the context of your research project, literature refers mainly to the text books and journal articles that you need to read to gain an understanding of the existing body of knowledge in your topic area. This includes the 'traditional or accepted wisdom', the main theories and concepts, and the research methodologies adopted by researchers in the field.

On the other hand, data and information are the statistics, facts, figures, materials and evidence about the organisation, industry, country or the context/subject of your research that you collect in order to answer your research questions. The data/information may

already be available (secondary research) or you may have to gather it yourself (primary research). Data can be obtained from both primary and secondary research. When such data is used in a particular way, it becomes information. Please note that data and information are not the same and not inter-changeable. Data is raw, unstructured bits and pieces of facts or ideas which require organization and ordering to convey meaning or become information. Information is organized data which conveys meaning, which if it can be used to solve problems, becomes knowledge.

4.6.2 Reasons for carrying out a literature review

Saunders *et al.* (2009, pp.58-59) state that there are two major reasons why you should review the literature. Firstly, the preliminary literature review that you should carry out at the proposal writing stage aims to help you generate and refine your research ideas, research objectives and research questions. Secondly, you will need to carry out the critical literature review as you start your research so that you gain a general awareness of the body of knowledge in your subject area and also to contextualize your topic in the global context, for you to know what has already been done in the field and the gaps to fill in.

Two other reasons for you to carry out a literature review are explained by Bryman and Bell (2011, pp.91-92) and Gall *et al.* (2006) cited in Saunders *et al.* (2009, pp.61-62) as follows.

- ✓ The review of the literature helps you to identify research that has already been carried out by other scholars so that you can avoid plagiarism (more on this later) by not replicating such research.
- ✓ The review of the literature also helps you to identify research approaches, research strategies and data collection and analysis techniques which you can use for your own research.

4.7 LITERATURE AND DATA/INFORMATION SOURCES

4.7.1 Literature sources

There are many literature sources that you can use for your business and management research project. These include journal articles, text books, theses, and conference papers (Saunders *et al.*, 2009, pp.68-75). Most of these sources are available in hard copy form, and increasingly on the internet.

The most likely sources you will use for your research however, are journal articles and text books because these are readily available e.g. through the ZCAS library or through online sources such as Google Scholar (more on this later). In the case of text books, you could actually already have them with you if you research on a topic that you have covered in one or more of your previous courses. Textbooks, official publications, and journals are preferred because they are peer-reviewed and authenticated.

4.7.2 Activity 4.1



At the end of ACTIVITY 3.1 you chose one of three topics (hopefully) as the most suitable, based on the good attributes of a topic. In respect of that topic (or any other that you may have decided upon), list the text books, journal articles and any other literature sources that you think are relevant for your literature review in the spaces below:

- a).....
- b).....
- c).....
- d).....
- e).....
- f).....
- g).....

4.7.3 Data/information sources

The data and/or information that you use in your research depend on your research questions and the context of your study. Such data may already have been collected for a different purpose (i.e. secondary data, further discussed in Section 5.11.9) or you may need to collect it yourself (primary data). The following are common sources of secondary data that you use in your research.

- ✓ The organisation's annual reports and website.
- ✓ The annual reports and website for the industry regulator of the research organisation.
- ✓ Government ministries and agencies such as the Central Statistics Office.

- ✓ International organisations such as the World Bank and IMF.
- ✓ Databases such as www.africanfinancials.com and www.bbcnews.org.

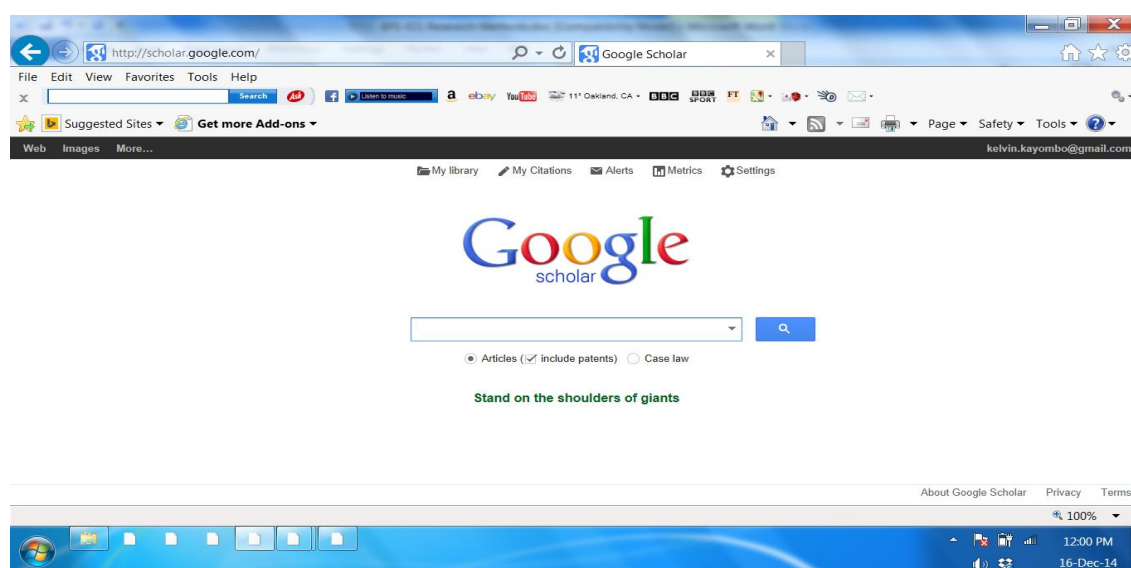
4.8 CARRYING OUT A LITERATURE SEARCH

You can search for hard copies of text books, journal articles and other literature sources by visiting the ZCAS library or other libraries in your area. In addition to hard copies, you are encouraged to use online resources, particularly databases that subscribe to high quality academic journals. Within ZCAS you can use the emerald database at www.emeraldinsight.com. This database can only be accessed when you are connected to the ZCAS local area network either wirelessly or otherwise. Outside ZCAS, you can use Google Scholar at www.scholar.google.com. Google Scholar is a fast and free resource that does not require any subscription or registration procedures. However, not all books and journal articles that Google Scholar turns up will be downloadable, as some may ask you to pay a subscription fee.

4.8.1 Simple search

Every database gives you an opportunity to carry out a simple search or an advanced search. A simple search enables you to use one search term only and does not have facilities to filter out irrelevant search results. A search term could be a key word in your topic title, research objective and research question or a theme you have generated for the literature review. Figure 4.1 below is a screen shot for the simple search screen in Google Scholar.

Figure 4.1 Google Scholar simple search screen shot (Source: Google Scholar, 2014)



4.8.2 Advanced search

The advanced search facility of any database or search engine avails you the opportunity to use several search terms to help you find literature that is very relevant to your research. This facility also has filters that assist in expanding the search (if search results are inadequate) or in narrowing the search to eliminate irrelevant results (if search results are too many). Figures 4.2 and 4.3 are screen shots of the advanced search feature in Google Scholar and Emeraldinsight respectively. Notice the various dialogue boxes that enable you to search using several search terms. Additionally, whereas there are three filters in Google Scholar i.e. where the search terms occur (anywhere in the document or in the title of the document), who the author is and when the article was published, Emeraldinsight (and indeed many other databases enable you to create as many filters as you wish. You are therefore encouraged to use the advanced search facility as this saves time. Consider also “Dogpile” search engine, which combines many search engines.

Figure 4.2 Google Scholar advanced search screen shot (Source: Google Scholar, 2014)

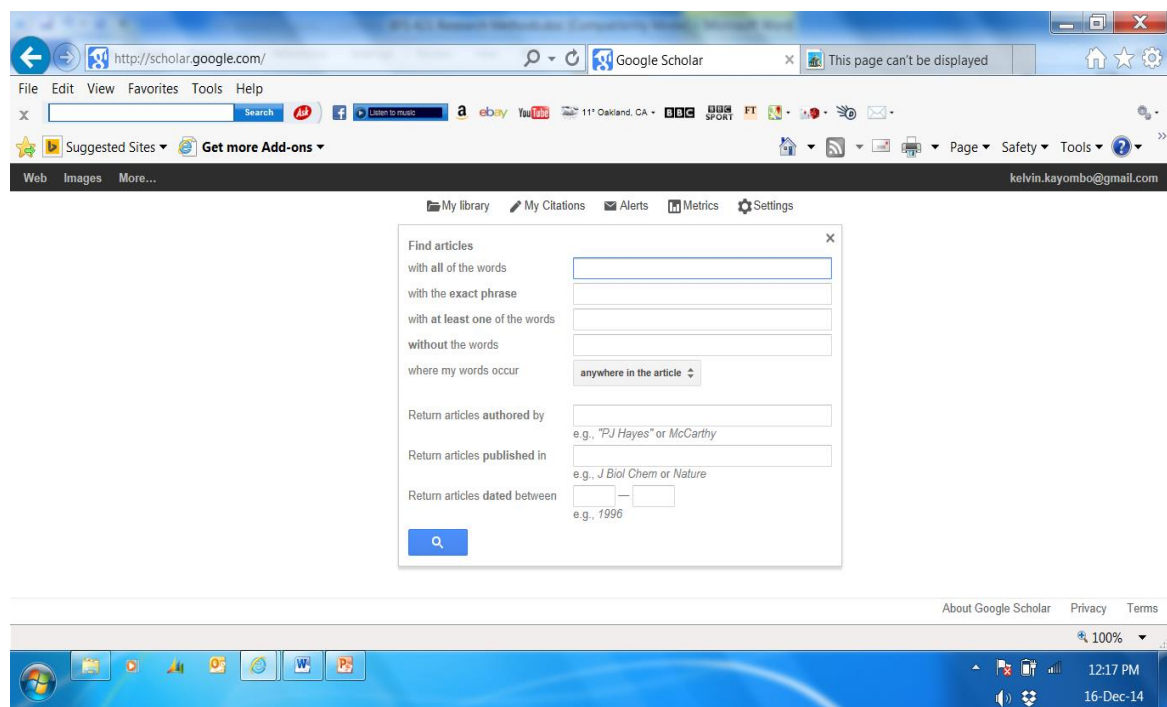
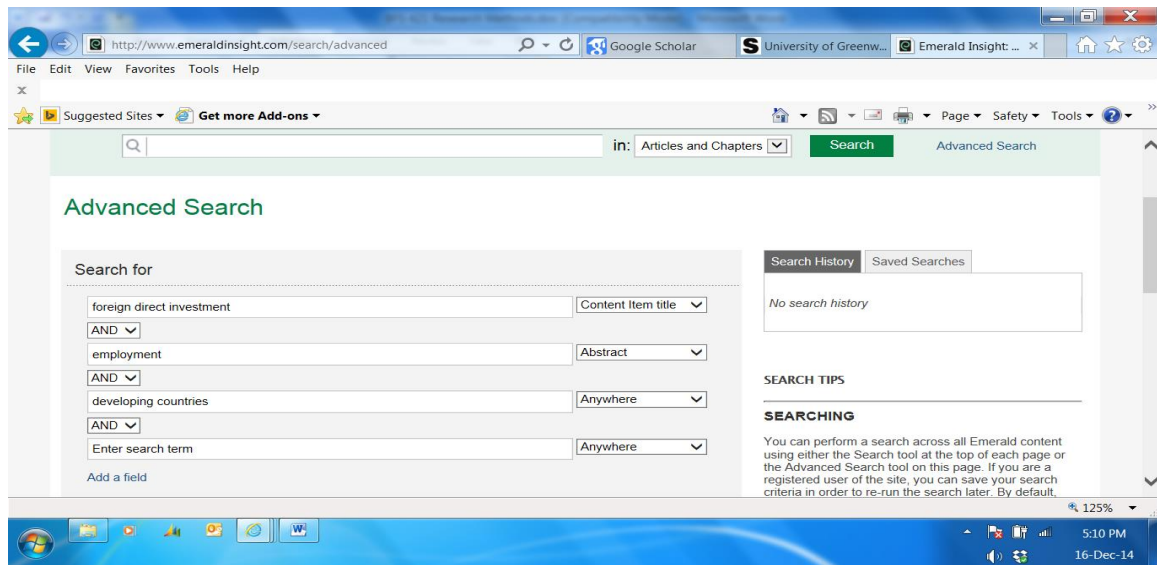


Figure 4.3 Emeraldinsight advanced search screen shot (Source: Emerald Group Publishing, 2014)



4.9 ACTIVITY 4.2



Let's suppose that you want to investigate the impact of corporate social responsibility activities on a company's performance. Go to www.scholar.google.com and type the search term 'corporate social responsibility' in the simple search screen and click on the search icon. Note the number of hits here.....(there were about 1.8 million hits at the time of writing).

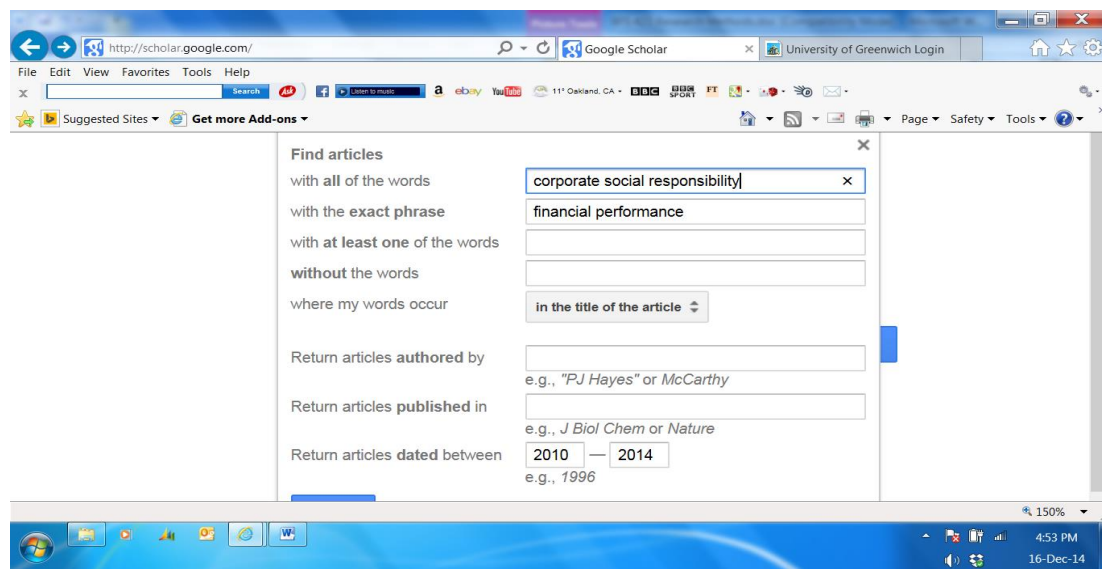
Now go back to the simple search screen and point your cursor on the downward arrow within the dialogue box (to the far right) to access the advanced search screen. Type the search words 'corporate social responsibility' and 'performance' in different dialogue boxes and note the number of hits this time.....(there were 1.48 million hits at the time of writing).

Try to use different search terms and the filters in advanced search mode e.g. 'corporate social responsibility' and 'financial performance' and ask Google Scholar to search in the topic title instead of anywhere in the article and insert a time range in the bottom filter (e.g. 2010 to 2014) and observe what happens (the advanced search screen should look like Figure 4.4 below. Note the number of hits here.....(there were only

266 at the time of writing). This number is now manageable and the articles are very relevant to the research topic.

Now use the advanced search facility to search for literature for the topic you identified in ACTIVITY 4.1. You need to generate search terms for this exercise and try different ones if the hits are not adequate.

Figure 4.4 Google Scholar advanced search screen shot with two filters activated (Source: Google Scholar, 2014)



4.10 CRITICALLY REVIEWING THE LITERATURE

4.10.1 Thematic review of the literature

Before you carry out a review of the literature, you need to identify the main themes that are relevant to your research questions because the literature review should be carried out and written thematically. Other themes may emerge as you review different literature sources. Do not review one literature source after another, but rather the themes contained in those sources. Tables 4.1 and 4.2 below illustrate the themes that could be used as a basis for reviewing and writing up the literature review on topics on child abuse and foreign direct investment (FDI) respectively.

Table 4.1 Examples of themes on a topic on child abuse

Theme	Articles Cited
A – extent and seriousness of the problem	1, 2, 5, 8, 9
B – current law, and policies	2, 4, 7, 8, 9, 10
C – poverty as a cause of child abuse	6, 7, 8, 9, 10
D – impact of child abuse	1, 5, 9
E – treatment and prevention of child abuse	3, 5, 9

Table 4.2 Examples of themes on a topic on foreign direct investment

Theme	Articles Cited
A – Types of FDI	1, 2, 5, 8, 9
B - Sources of FDI	2, 4, 7, 8, 9, 10
C – Benefits of FDI	6, 7, 8, 9, 10
c (i) Employment creation	7, 10
c (ii) Poverty reduction	1, 7, 8
c (iii) Balance of payments	2, 5, 9
c (iv) Economic development	
D – Policies for attracting FDI	3, 5, 9
E – Adverse effects of FDI on host countries	7, 9, 10

4.10.2 Contents of the critical review

Your literature review should be based on your research objectives and research questions, and it should be the basis for the data collection and analysis that follows (Saunders *et al.*,

2009, p.66). They contend that the review should ‘evaluate the research that has already been undertaken in the area of your research project, show and explain the relationships between published research findings and reference the literature in which they were reported’. The literature review should therefore include the key academic theories in your research field, a history of the development of the field including the very latest research findings, literature from a wide range of geographical sources (e.g. Europe, America, Asia, Africa etc.) and adequate citations and referencing (empirical research which is time and spatial in dimension).

4.10.3 Critiquing the literature

Saunders *et al.* (2009, p.64) advise that as you critically review the literature, ‘you need to read the literature about your research topic with some skepticism and be willing to question what you read’. They cite Mingers (2000, pp. 225-226) regarding the specific issues that you should question or critique, i.e. the accepted/conventional wisdom or ‘critique of tradition’; the overriding stance that the literature depicts or ‘critique of authority’; and the knowledge and information you are reviewing (e.g. validity, reliability and generalizability of the previous research) or the ‘critique of objectivity’. Being critical is also about how you make reasoned judgements and how you demonstrate your argumentative skills in your writing i.e. how convincing you are or the ‘critique of rhetoric’.

4.11 IN-TEXT CITATIONS AND REFERENCING

4.11.1 Plagiarism

Plagiarism is defined as ‘presenting the work and ideas of other people and passing them off as your own, without acknowledging the original source of the ideas used’ (Easterby-Smith *et al.*, 2008, p.50 cited in Saunders *et al.*, 2009, p.97). The actual offences include outright theft (buying, copying or submitting another student’s work), submitting another person’s paper as your own, paraphrasing (writing in your own words) material without acknowledging the source and presenting material word for word with proper citations but failing to put the material either in quotation marks or indented in the text in a different font size. Remember that you are allowed to (and you should) use other people’s work in your own research as long as you do not pass it off as your own.

Plagiarism is a very serious academic offence. If proven, penalties include reduction in marks gained, reprimand, failure of the specific course-work, failure of the entire course or expulsion from the university.

You should be aware that plagiarism is very easy to detect. As you know, ZCAS uses software called TurnItIn (see www.turnitin.com) to scan all your electronically-submitted course-work such as assignments, the research proposal, and the dissertation itself. To avoid plagiarism, do the following:

- a) When you copy another person's (or organisation's) work verbatim, put the copied material in quotation marks and cite the source (or vice versa). However, direct quotes should be restricted to definitions and key concepts only. It is not advisable to copy large chunks of text verbatim as this suggests that you have not understood the material or you are lazy and you indulge in copy and paste!
- b) Paraphrase material that you use and cite the source. This means expressing what you have read in your own words without changing the meaning. As you paraphrase, please change both the sentence and paragraph structure i.e. the sequence and number of words and sentences should be different. Do not be tempted to copy sentences and then simply change a few words here and there because TurnItIn is able to detect this inadequate paraphrasing.

4.11.2 Harvard and APA Referencing System

The Harvard Referencing System is the official referencing system for your programme of study. Like the APA (American Psychological Association) style, the Harvard style is a Name and Date (author and year) referencing system that is commonly used in business and management academic writing. These two styles are very similar, allow for flexibility in how they are applied and many authors use them simultaneously. For example, Saunders *et al.* (2009) and Bryman and Bell (2011) have used the Harvard style strictly, while a combination of the two styles has been adopted in this module. The important thing is consistency of application.

Both the Harvard and APA referencing styles require the in-text citation to show the name of the author (surname only) and year of publication. The references list should include full details of all sources that you cited in the text. These details are the name and initials of the author, year of publication, title of source, and publication details such as city of

publication, and publisher's name. In order to avoid plagiarism, you should ensure that only sources (and all of them) that you have cited in the text are included in the references list. Other sources used but not cited as in-text may be put separately as bibliography.

4.12 REQUIRED READING 4.1



Now read Chapter 3 in Saunders *et al.* (2009) and/or Chapter 4 in Bryman and Bell (2011).

4.13 ACTIVITY 4.3



Now attempt the self-check and discussion questions in Saunders *et al.* (2009, pp.99 – 93).

4.14 SUMMARY



This unit has taken you through the stages of carrying out a critical review of the literature and introduced you to the associated key terms and concepts. You need to distinguish between the terms 'literature' and 'data'/'information' as used in research and be aware of the sources of both. The unit has emphasised that you should use the literature review not only to clarify your research objectives and questions but also as a theoretical framework for the latter stages of your study.

Before you can review the literature, you need to find it first. Books and journals can be accessed from libraries (e.g. the ZCAS library) and online sources (e.g. Google Scholar and Emeraldinsight). The advanced search facility is an important search tool when using online sources because it has filters that help you to focus your search.

The review of the identified literature should be thematic and involve evaluating and weighing the findings of different researchers against each other. The unit has encouraged you to critique the traditional wisdom, the dominant view and the methods adopted by other researchers. The language that you use in writing up your review should also be convincing.

As you write your literature review, bear in mind the need to acknowledge other authors' work to avoid plagiarism. Plagiarism is a serious academic offence, equivalent to intellectual theft and dishonesty. Adopting the Harvard Referencing System as required by this programme will help you to provide an academic piece of work that is suitably evidenced.

Having reviewed the literature and identified a gap that your research can fill, you now need to start the research proper. This naturally involves you collecting and analysing data. The next unit on research methodology discusses the conceptual and practical issues that you need to address.

4.15 ASSIGNMENT 1 SUBMISSION



You should now attempt and submit the first assignment. Assignment 1 is a critical appraisal of a literature review section. This assignment was issued at commencement of the course and is due during the second week of April (if you are taking this course during the January to June semester) or second week of October (if you are taking this course during the July to December semester). Verify the actual submission deadline on Moodle or the academic calendar or contact your course lecturer.

Note that this assignment accounts for 25% of your total marks for this course. The word count should be 1,000 words +/- 10%. Failure to meet this word count attracts penalties. You should use the Times New Roman font, size 12, line spacing of 1.5 and justify the text. You should submit your assignment through TurnItIn. Do not e-mail the assignment to the lecturer.

5.0 UNIT FOUR: RESEARCH METHODOLOGY



5.1 INTRODUCTION

The research methodology is the action plan for implementing your research. It describes the conceptual framework and the specific actions that you should take to answer your research questions and achieve your research objectives. The conceptual issues that you need to grasp include your research paradigm, whether your study is exploratory,

explanatory, descriptive or a combination of any of these types; whether your research is qualitative or quantitative or both; and whether to adopt the inductive or deductive approach in relating theory to data. The practical or operational issues include your choice of research strategy, sampling techniques, and data collection instruments. This unit introduces you to these aspects of your research. It also justifies reasons for using certain research approaches.

5.2 AIM



The aim of this unit is to equip you with the knowledge and tools that you need to implement your research project.

5.3 OBJECTIVES



By the end of this unit, and having completed the prescribed readings and activities, you should be able to:

1. Explain types of research e.g. descriptive, exploratory and explanatory studies.
2. Distinguish between quantitative and qualitative research methods.
3. Distinguish between inductive and deductive research approaches.
4. Explain the different research strategies including experimental, survey, case study, grounded theory, action research, and cross-sectional research.
5. Explain and evaluate data collection tools including sampling methods, observation, interviews, questionnaires, document review, and secondary data.
6. Distinguish between validity, reliability and generalizability, and how triangulation can be used to enhance these aspects of research.

5.4 TIME REQUIRED



This is the longest unit in this module as it covers a wide range of issues. You should spend a minimum of 40 hours on this unit. This time should be spent

on reading the contents of this unit, reading the prescribed reading [Section 4.3 of Chapter 4 and Chapters 5, 7, 8, 9, 10 and 11 in Saunders *et al.* (2009)] and on attempting Activity 5.1 – 5.3 as well as the self-check and discussion questions in the prescribed readings itemized above.

5.5 REFLECTION



You have decided what your research is about. You have reviewed the literature, clarified your research objectives and research questions. Now you need to get your hands ‘dirty’ by delving into the research proper. What should you do? Where will the data that you will need to answer your research questions come from? How are you going to collect that data?

5.6 RESEARCH PARADIGM

Saunders *et al.* (2009, p.118) define the term ‘paradigm’ as used in business and management research as ‘a way of examining social phenomena from which particular understanding of the phenomena can be gained and explanations attempted’. Your research paradigm is a reflection of your epistemology i.e. what you consider as acceptable knowledge. On the one hand you could only accept ‘facts’, ‘reality’ and ‘objective’ knowledge, while on the other, you could accept knowledge derived from ‘feelings’ and ‘attitudes’ and tolerate ‘subjectivity’ in studying social phenomena. Epistemology deals with sources of knowledge and ways of knowing, what is usually referred to as theory of knowledge.

Bryman and Bell (2011, pp.15-20) and Saunders *et al.* (2009, pp.112-116) assert that positivism and phenomenology are the two main and opposing epistemological considerations for business and management research. Saunders *et al.* (2009, p.113) state that a positivist study involves using existing theory to develop hypotheses for testing. Such a study ‘will be concerned with facts rather than impressions’ and ‘such facts are consistent with the notion of “observable social reality” similar to that employed by the physical and natural scientists’ (p.114) or what is popularly called the scientific method which was developed by Thomas Kuhn, Roger Bacon, Karl Popper, Lakatos, among others. Cohen *et al.* (2007, p.7) add that the positivist researcher views knowledge as being hard, objective and tangible, and therefore avoids getting involved with the research

subjects. Research carried out in this paradigm is usually quantitative in nature and deductive in relating theory to data.

In contrast, a phenomenological epistemology position is ‘anti-positivist’ (Cohen *et al.*, 2007, p.21); and requires the researcher ‘to enter the social world of our research subjects and understand their world from their point of view’ (Saunders *et al.*, 2009, pp. 115 – 116). Research carried out in this paradigm is usually qualitative in nature and inductive in relating theory to data.

The choice of research paradigm depends on your research objectives and questions. The positivism and phenomenological paradigms can be considered to be extreme ends of a continuum; consequently, some research projects can be thought of as being more positivist or more phenomenological than others. However, your research is either one or the other, not both. In other words the two paradigms are mutually exclusive.

5.7 DESCRIPTIVE, EXPLORATORY AND EXPLANATORY STUDIES

The type of research that you undertake depends on the purpose for which the research is carried out, which in turn is determined by your research questions and objectives. There are three types of research, namely exploratory, explanatory and descriptive. According to Saunders *et al.* (2009, pp. 139 – 140) the purpose of an exploratory study is to ‘clarify your understanding of a problem, such as if you are unsure of the precise nature of the problem’. This can be done by searching the literature, interviewing experts in the subject area or carrying out focus group discussions. Such studies are intended to answer ‘why’ and ‘how’ questions.

Explanatory studies seek to establish causal relationships between variables (Saunders *et al.*, 2009, p.140). You use such studies to answer ‘what’ type of questions.

Descriptive research is designed ‘to portray an accurate profile of persons, events or situations’ (Robson, 2002, p.59 cited in Saunders *et al.*, 2009, p.140). This type of research is likely to be more factual and objective, but lacks evaluation and synthesis. Accordingly, it is unlikely that your research will be solely descriptive as it may not satisfy the assessment requirements at the final level of an undergraduate programme. Instead, you are likely to use this type of research as a precursor to the more cognitively demanding exploratory or explanatory studies.

5.8 DEDUCTIVE AND INDUCTIVE RESEARCH

There are two main approaches that you adopt as you conduct your research. The inductive approach involves you collecting data and developing theory as a result of the data analysis (Saunders *et al.*, 2009, p.125; Adams *et al.*, 2007, p.29) and it is normally associated with qualitative research. On the other hand, if you adopt the deductive approach, then you would develop a theory and hypotheses and design a research strategy to test your hypotheses (Saunders *et al.*, 2009, pp.124 – 125; Adams *et al.*, 2007, pp.29 – 30). This approach is normally associated with quantitative research. In practice however, you can make use of both deduction and induction to benefit from the best of both approaches (Saunders *et al.*, 2009, p.127).

5.8.1 Required reading 5.1



Now read Saunders *et al.* (2009, pp.124-128).

5.9 QUANTITATIVE AND QUALITATIVE RESEARCH

Bryman and Bell (2011, pp.26-27) consider quantitative research to be a strategy ‘that emphasizes quantification in the collection and analysis of data’. Such a strategy is considered to be scientific in nature and is associated with the deductive approach to research. Questionnaires are examples of data collection techniques you would use for this method of carrying out research, while graphs or statistical analysis would be an example of analysis techniques that you employ (Saunders *et al.*, 2009, p.151). You may also do tests of significance by interpreting results, using known tests such as chi-square.

On the other hand, Bryman and Bell (2011, p.27) state that qualitative research ‘emphasises words rather than quantification in the collection and analysis of data’. Such a strategy is considered to be interpretive in nature and is associated with the inductive approach to research. Interviews are examples of data collection techniques or instruments you would use for this method of carrying out research; while categorizing data would be an example of analysis techniques you employ (Saunders *et al.*, 2009, p.151).

5.9.1 Required reading 5.2



Now read Saunders *et al.* (2009, pp.151-155) and/or Bryman and Bell (2011, pp.26-28).

5.9.2 Activity 5.1



Here is a short exercise to help you begin to appreciate the differences between qualitative and quantitative data.

- a) Choose a small community with which you are familiar. It could be a department in your organisation, a community group to which you belong, or even your extended family.
- b) Think about all the ways in which you could describe this community using quantitative methods (things like age, gender, height and hair colour etc.).
- c) Select three of these features; gather data for all the members of the group. Now see what you can do with the data. Look at things like the range across the group and the average for the group. Can you see any relationships between the features, for example between height and gender, or between age and numbers of children/grandchildren? In research terms, these features are called variables.
- d) Now consider the views that your group of people may have about a relevant issue. This might be a political issue, an organisational issue, a social issue or something which is in the news. Choose one of these issues and talk to a few members of your group about it. By doing this you are using a qualitative method.
- e) Reflect on both pieces of research. What did your data tell you? How could you present your findings to your group in order to give them an overall impression of what you found?
- f) In what circumstances would you use quantitative data collection methods, and in what circumstances would you use qualitative methods?
- g) What has your brief experiment told you about ways of collecting data, deciding who to include in your research (your sample), and ways of recording your findings, ways of analyzing your findings and ways of presenting your findings to another person or other people?

5.10 RESEARCH STRATEGIES

5.10.1 Case study

Yin (1981) states that a case study is distinguishable from other research strategies because it examines ‘a contemporary phenomenon in its real-life context’. Robson (2002, p.178) gives a more comprehensive definition of a case study as ‘a strategy for doing research which involves an empirical investigation of a particular contemporary phenomenon within its real life context using multiple sources of evidence’. Although you can use this strategy to answer ‘what’ and ‘how’ questions, it is more suitable for answering ‘why’ questions. Consequently, you are more likely to employ case study strategy if your research is exploratory or explanatory (Saunders *et al.*, 2009, p.146).

The case study strategy is one of the strategies you are likely to adopt for your research. This is because, at undergraduate level, the scope of your research is likely to be narrow e.g. based on one organisation, community or industry.

5.10.2 Survey

A survey is a large scale data collection strategy that is associated with quantitative research and the deductive approach. This strategy is commonly used to answer who, what and how questions and is often used for exploratory and descriptive studies (Saunders *et al.*, 2009, p.144). Survey results are generally authoritative, easier to understand and the standardized data generated is suitable for comparison. The most common data collection instrument you would use in a survey is the questionnaire; this limits the amount of data that you can collect because there is a limit to the number of questions that your respondents can be expected to answer.

5.10.3 Archival research

Saunders *et al.* (2009, p.150) state that archival research ‘makes use of administrative records and documents as the principal source of data’. Even though it is possible for you to use this strategy (and many researchers actually do use it, albeit in conjunction with another strategy), you need to be aware that some data might be missing or you may not have access to it. You may therefore need to use this strategy to supplement other research strategies.

5.10.4 Experiment

Saunders *et al.* (2009, p.144) state that this strategy is used if you want to establish whether there is a causal relationship between two or more variables, the size of such links and which variables are more important in the causal relationship. Experiments are often used to answer ‘why’ and ‘how’ questions in exploratory and explanatory research in the natural sciences. Accordingly, it is highly unlikely that you will use this strategy in your research in business and management.

5.10.5 Grounded theory

Grounded theory is a strategy in which no theoretical framework is used as a basis for the research; instead the theory is developed from the data, tested and then more data is collected and this iterative process repeated further (Saunders *et al.*, 2009, p.148-149). It is therefore highly unlikely that you will have the resources to use such a strategy.

5.10.6 Action research

Action research is defined as a strategy in which the researcher and the research participants work together to diagnose a problem and find a solution (Bryman and Bell, 2011, p.413). The researcher becomes engrossed in the research to gain a deeper understanding of the situation and people in the organisation. Sociologists, anthropologists, educators and educationists, psychologists, and criminal investigators use this method of research.

5.10.7 Required reading 5.3



Now read Saunders *et al.* (2009, pp.141-151).

5.10.8 Activity 5.2



Consider the research topic you chose in Activity 3.1 and the ROs and RQs you designed in Activity 3.2. Which strategy or strategies do you think will help you answer your RQs more effectively? Write down your proposed research strategy here.....

.....

5.11 LONGITUDINAL AND CROSS-SECTIONAL RESEARCH

Saunders *et al.* (2009, p.155) refer to a cross-sectional study simply as ‘the study of a particular phenomenon (or phenomena) at a particular time’. Such a research does not require mapping changes in the variables over time and hence the research is a ‘snapshot’ at a given time (Bryman and Bell, 2011, pp.53-54). Most, if not all, undergraduate research, including your own study is likely to be cross-sectional.

A longitudinal design, on the other hand, is usually carried out over a long time horizon to map or study change and development. Such a time horizon is less suitable for your research project due to time and cost constraints (Bryman and Bell, 2011, p.57; Saunders *et al.*, 2009, p.155).

5.11.1 Required reading 5.4



Now read Saunders *et al.* (2009, pp.151-155) and/or Bryman and Bell (2011, pp.26-28).

5.12 DATA COLLECTION TECHNIQUES AND INSTRUMENTS

5.12.1 Sampling

Sampling is the application of data collection procedures to less than 100% of all possible research participants. Sampling is normally required because you may not have sufficient resources in terms of time, money and access to interview or administer a questionnaire (or any other data collection instrument) to everyone (or everything) in your population of interest or universe of discourse (Saunders *et al.*, 2009, p.210). You would therefore administer your data collection instrument to some chosen members of the parent population (participants only).

There are two main categories of the techniques involved in selecting a sample or a representative group of the parent population. Probability sampling techniques such as random, systematic, stratified and cluster sampling are statistical in nature and give each item or sampling unit in your population an equal chance of being selected. These techniques are associated with survey and experimental strategies; and because the sample selected has a high chance of representing the characteristics of the population, you are able to make conclusions on the population based on the sample results and findings. This avoids bias as sample statistics derived approximate the population parameters.

Conversely, non-probability sampling techniques such as quota, purposive, snowball, self-selection and convenience sampling involve subjective judgement in selecting the sample (Saunders *et al.*, 2009, p.233). The sample selected may therefore not represent characteristics of the population from which it was selected. Consequently, non-probability sampling strategies preclude generalizing sample results to the population on statistical grounds (Cohen, Manion and Morrison, 2007, p. 113). However, it is still possible to generalize the findings to theory (Yin, 2010, pp.99-100). Yin refers to this kind of generalization as ‘analytic generalization’ (98-102).

5.12.2 Required reading 5.5



Now read Chapter 7 in Saunders *et al.* (2009) and/or Chapter 7 in Bryman and Bell (2011).

5.12.3 Interviews

Interviews are a common data collection tool that can assist you to collect valid and reliable data. Saunders *et al.* (2009, pp.318-321) posit that interviews can be categorised into three groups depending on your research question, research objectives, the purpose of your research and the research strategy that you have adopted. Structured interviews or interviewer-administered questionnaires are normally used to collect quantitative data using standardised questionnaires in descriptive research.

Semi-structured interviews are non-standardised in that questions may vary between interviews, with some actually being omitted (Saunders *et al.*, 2009, pp.320-321). Such interviews are suitable for explanatory research and to a lesser extent exploratory research. Both open-ended and closed questions may be asked during this type of interview. If your research is qualitative in nature then this is the most likely type of interview that you will conduct.

Unstructured or in-depth interviews are informal, with no pre-determined set of questions (Saunders *et al.*, 2009, p.321). Such interviews are only suitable for exploratory research and their lack of structure makes them unsuitable for your research project.

5.12.4 Required reading 5.6



Now read Chapter 10 in Saunders *et al.* (2009).

5.12.5 Questionnaires

Questionnaires are data collection instruments that are mostly used in a survey strategy (Saunders *et al.*, 2009, p.360) and are normally associated with descriptive and explanatory research. Research participants are asked the same questions and may be required to record their own answers. Questionnaires can be self-administered (by the respondent) or interviewer-administered. Examples of self-administered questionnaires include internet and postal questionnaires while telephone questionnaires and structured interviews are examples of interviewer-administered questionnaires.

Saunders *et al.* (2009, pp.371-372) assert that reliability of a questionnaire largely depends on questionnaire design, structure and rigour of pilot testing. They accordingly recommend the use of software in designing the questionnaire and pilot testing, *inter alia*, to achieve reliability. Bryman and Bell (2011, pp.262-263) and Cohen *et al.* (2007, pp.341-342) also strongly recommend carrying out a pilot study before administering a questionnaire. Pre-coding and pilot testing a questionnaire also enhance its validity and practicability.

5.12.6 Required reading 5.7



Now read Chapter 11 in Saunders *et al.* (2009).

5.12.7 Observation

According to Saunders *et al.* (2009, p.288) observation as a data collection instrument involves ‘the systematic observation, recording, description, analysis and interpretation of people’s behaviour’. There are two main types of observation namely participant observation and structured observation.

Observation is commonly used in sociology and anthropology; but it is hardly used in business and management research (Saunders *et al.*, 2009, pp.289-290). It is therefore unlikely that you will use this data collection instrument in your research.

5.12.8 Required reading 5.8



Now read Chapter 9 in Saunders *et al.* (2009).

5.12.9 Secondary data

As stated in Section 4.7.3 the data/information that you use in your research may already have been collected by other parties (secondary data) for their own purposes or you may have to collect the data yourself (primary data). Alternatively, and more common, you could use both primary and secondary data in your research project. Generally speaking, primary data are more expensive and time-consuming to gather than secondary data; therefore you should consider availability of secondary data before you decide to carry out primary research.

Secondary data can be qualitative or quantitative and are normally used in descriptive and explanatory research (Saunders *et al.*, 2009, p. 258). There are numerous sources of secondary data that you can consult. Examples include organisations' databases, survey reports, communications and websites, reports and minutes of committees, journals, newspapers, diaries, interview transcripts, media accounts, Government publications, surveys and census reports, and industry statistics and reports (p.259). The specific source that you use depends on your research questions.

5.12.10 Required reading 5.9



Now read Chapter 8 in Saunders *et al.* (2009).

5.12.11 Activity 5.3



Consider the ROs and RQs you designed in Activity 3.2 and the research strategy or strategies you proposed in Activity 5.2. Which data collection techniques do you think will help you answer your RQs more effectively? Write down your proposed data collection instrument(s) here.....
.....

5.13 RELIABILITY, VALIDITY AND GENERALISABILITY (GLOBAL CONTEXT) OF RESEARCH FINDINGS

Reliability

Saunders *et al.* (2009, p.156) consider reliability to be ‘the extent to which your data collection techniques or analysis procedures will yield consistent findings’. Reliability of research findings can be enhanced by using various techniques. These include the use of focus group and interview guides to eliminate observer error (p.157); provision of participants with a list of the interview themes from the interview guides prior to the event (p.328); taking notes during the interview/discussion (p.334); and adapting existing questions.

Saunders *et al.* (2009, pp.371-372) assert that reliability in survey research employing a questionnaire largely depends on questionnaire design, structure and rigour of pilot testing. They accordingly recommend that you use software in designing the questionnaire and that you pilot test such a questionnaire. Bryman and Bell (2011, pp.262-263) and Cohen, Manion and Morrison (2007, pp.341-342) also strongly recommend that you carry out a pilot study before administering a questionnaire. The questionnaire should be logically designed.

Validity

Validity is an important criterion for evaluating the credibility of your research findings. It is concerned with the integrity of your research findings, i.e. whether your findings really reflect the reality on the ground (Bryman and Bell, 2011, pp.42-43; Hair *et al.*, 2010, pp.7-8; Saunders *et al.*, 2009, p.157). Bryman and Bell (2011, p.397) and Yin (2010, p.81) posit that validity of qualitative research findings can be enhanced through triangulation. Saunders *et al.* (2009, p.146) define triangulation as ‘the use of different data collection techniques within one study in order to ensure that the data are telling you what you think they are telling you’. Bryman and Bell (2011, p.397) have a similar view when they state that triangulation ‘entails using more than one method or source of data in the study of social phenomena’.

Generalisability

Generalisability is the extent to which research findings from one study can be applied to other settings (Saunders *et al.*, 2009, p.158). Generalisability is generally accepted as a problem in qualitative research due to small unrepresentative samples that are usually selected for the study (Bryman and Bell, 2011, pp.408-409). However, it is still possible to generalize the sample research findings to theory, even if generalization cannot be made to the population (Bryman and Bell, 2011, pp.408-409; Yin, 2010, pp.99-100; Saunders *et al.*, 2009, p.213; Cohen, Manion and Morrison, 2007, p. 113).

5.13.1 Required reading 5.10



Now read Saunders *et al.* (2009, pp.156-159).

5.14 SUMMARY



This unit on the research methodology has introduced you to the conceptual and operational issues that you need to address in order to answer your research questions. The theoretical issues include deciding your research paradigm i.e. positivism or phenomenology and deciding the type of your research i.e. whether it is exploratory, explanatory, descriptive or a combination. At undergraduate level, your research is more likely to be explanatory than the other two. The other conceptual issue you need to address is how you relate theory to data. If you design your research to collect data in order to test hypotheses developed from theory, then yours is the deductive approach. The converse, i.e. inductive approach, is where you develop theory from data. The deductive approach is associated with quantitative research (collection and analysis of numerical data), while the inductive approach is affiliated with qualitative research (collection and analysis of non-numerical data).

The operational issues that you need to resolve include choice of appropriate research strategy, sampling techniques, data collection instruments and credibility of your research. As a business and management researcher at undergraduate level, you are more likely to choose among the case study, survey or archival research strategies than experiment (more suited to science research), grounded theory (resource constraints) or action research

(access and resource constraints) strategies. The choice of your research strategy should be determined by your research objectives and research questions.

In order to answer your research questions, you need to collect either primary data (costly and time consuming) or secondary data (suitability and access issues) or both. The most suitable data collection instruments for your primary research are likely to be interviews (in particular semi-structured interviews) and questionnaires. Whichever data collection instrument you choose, think about the reliability of your techniques and validity of your findings. Adopting existing interview questions and using an interview schedule can enhance reliability of your interview; while using software to design a questionnaire that is then pilot-tested increases credibility of your questionnaire.

5.15 ASSIGNMENT 2 SUBMISSION



You should now attempt and submit the second assignment. Assignment 2 is a critical appraisal of a research methodology section. This assignment was issued during the first week of April (if you are taking this course during the January to June semester) or first week of October (if you are taking this course during the July to December semester) and it is due during the second week of May (if you are taking this course during the January to June semester) or second week of November (if you are taking this course during the July to December semester). Verify the actual submission deadline on Moodle or the academic calendar or contact your course lecturer.

Note that this assignment accounts for 25% of your total marks for this course. The word count should be 1,000 words +/- 10%. Failure to meet this word count attracts penalties. You should use the Times New Roman font, size 12, line spacing of 1.5 and justify the text. You should submit your assignment through TurnItIn. Do not e-mail the assignment to the lecturer.

6.0 UNIT FIVE: DATA PRESENTATION AND ANALYSIS

6.1 INTRODUCTION



The manner in which you process and present your data and the techniques that you use to analyse it depend on whether the data is quantitative or qualitative.

Data processing, presentation and analysis can be carried out manually, with the aid of a spread-sheet or, more likely, with the aid of computer software. You covered quantitative data presentation and analysis techniques in the BAE 132 Introduction to Quantitative Methods course, so the relevant content is reproduced in Sections 6.8 and 6.9 below for you to revise. The most common techniques for analysing qualitative data are thematic and content analysis.

6.2 AIM



The aim of this unit is to equip you with the knowledge and skills required to process, present, and analyse your research data to enable you to answer your research questions.

6.3 OBJECTIVES



By the end of this unit, and having completed the prescribed readings and activities, you should be able to:

1. Explain qualitative data analysis including techniques such as thematic and content analysis.
2. Explain quantitative data analysis including presentation of data using graphs and charts.
3. Demonstrate an understanding of descriptive statistics such as distribution, dispersion and central tendency of the variable.
4. Distinguish between causation and correlation.
5. Explain inferential statistics including normal and non-normal distributions, hypothesis testing, confidence intervals, type I and II errors.
6. Test for significant relationships.
7. Use common data processing and analysis software such as Strata, Matlab, R, Eviews, Nvivo, Atlas.ti and SPSS.

6.4 TIME REQUIRED



You should spend a minimum of 30 hours on this unit. This time should be spent on reading the contents of this unit, reading the prescribed reading [Chapters 12 and 13 in Saunders *et al.* (2009) and/or Chapters 14 and 22 in Bryman and Bell (2011)] and on attempting the self-check and discussion questions in the prescribed readings itemized above.

6.5 REFLECTION



You have decided what your research is about. You have reviewed the literature and clarified your research objectives and research questions. You are clear about how your data are to be collected; but then that data needs to be presented and analysed so that you can answer your research questions. What should you do? Should you present your data on a chart, graph or table? What software, if any, should you use to help you with the presentation and analysis of your research findings?

6.6 QUALITATIVE DATA ANALYSIS

Qualitative data analysis ranges from simple categorisation of data to complex processes where relationships between data categories are identified and analysed (Saunders *et al.*, 2009, p.480). The analysis can be carried out manually or with the aid of a spread-sheet or more commonly these days, by using computer-aided qualitative data analysis software (CAQDAS) such as Nvivo and Atlas.ti. The two main techniques for analysing qualitative data are thematic analysis and content analysis.

Thematic analysis is one of the most common ways of approaching qualitative data analysis. It involves identifying themes in the data to find an analytic path within the voluminous data generated by qualitative research (Bryman and Bell, 2011, pp.571 - 572). Ryan and Bernard (2003) view themes as ‘abstract (and often fuzzy) constructs that link not only expressions found in texts but also expressions found in images, sounds and objects’. Bryman and Bell (2011, p.297) state that ‘when the process of coding is thematic,

a more interpretative approach needs to be taken'. This facilitates identification of latent content in addition to manifest content that characterize categorisation of specific words.

Content analysis is defined by Bryman and Bell (2011, p. 291) as 'An approach to the analysis of documents and texts that seek to quantify content in terms of predetermined categories and in a systematic and replicable manner'. This technique enables a researcher to discern the importance of values or concepts by measuring the frequency with which they occur.

6.6.1 Required reading 6.1



Now read Chapter 13 in Saunders *et al.* (2009).

6.6.2 Activity 6.1



Attempt the self-check and discussion questions in Chapter 13 in Saunders *et al.* (2009).

6.7 QUANTITATIVE DATA ANALYSIS

Quantitative data analysis techniques range from 'creating simple tables or diagrams that show the frequency of occurrence and using statistics such as indices to enable comparisons, through establishing statistical relationships between variables, to complex statistical modeling (Saunders *et al.*, 2009, p.414)'. You can carry out this analysis by using a spreadsheet such as Microsoft Excel or, more commonly, by using statistical packages such as SPSS. The presentation and analysis of quantitative data is discussed in greater detail in Sections 6.8 and 6.9 below. The material in these sections is reproduced from the BAE 132 Introduction to Quantitative Methods course that you covered in the first year of your studies (or an equivalent course if you were exempted).

6.8 DESCRIPTIVE STATISTICS

6.8.1 Frequency distributions

Definitions

Raw Data

Data collected in original form.

Frequency

The number of times a certain value or class of values occurs.

Frequency Distribution (Table)

The organization of raw data in table form with classes and frequencies.

Categorical Frequency Distribution

A frequency distribution in which the data is only nominal or ordinal.

Ungrouped Frequency Distribution

This is a frequency distribution of numerical data. The raw data or scores is not grouped.

Grouped Frequency Distribution

This is a frequency distribution where several numbers are grouped into classes.

Class Limits

Class limits separate one class in a grouped frequency distribution from another. The limits could actually appear in the data and have gaps between the upper limit of one class and the lower limit of the next.

Class boundaries

They separate one class in a grouped frequency distribution from another. The boundaries have one more decimal place than the raw data and therefore do not appear in the data. There is no gap between the upper boundary of one class and the lower boundary of the next class to form a continuous rather than a discrete distribution. The lower class boundary is found by subtracting 0.5 units from the lower class limit and the upper class boundary is found by adding 0.5 units to the upper class limit.

Class Width

It is the difference between the upper and lower boundaries of any class. The class width is also the difference between the lower limits of two consecutive classes or the upper limits

of two consecutive classes. It is not the difference between the upper and lower limits of the same class.

Class Mark (Midpoint)

It is the number in the middle of the class. It is found by adding the upper and lower limits of a particular class and dividing by two. It can also be found by adding the upper and lower boundaries and dividing by two.

Cumulative Frequency

It is the number of values or scores less than the upper class boundary for the current class. This is a running total of the frequencies.

Relative Frequency

It is the frequency of a class divided by the total frequency. This gives the percentage of values falling in that class.

Cumulative Relative Frequency (Relative Cumulative Frequency)

It is the running total of the relative frequencies or the cumulative frequency divided by the total frequency. It gives the percentage of the values which are less than the upper class boundary.

Histogram

A histogram is a graph which displays the data by using vertical bars of various heights to represent frequencies. The horizontal axis can be the class boundaries, the class marks, or the class limits.

Frequency Polygon

It is a line graph. The frequency is placed along the vertical axis and the class midpoints are placed along the horizontal axis. These points are connected with lines from the origin to an imaginary midpoint of an assumed class after the last actual class.

Ogive

It is a frequency polygon of the cumulative frequency or the relative cumulative frequency. The vertical axis shows the cumulative frequency or relative cumulative

frequency. The horizontal axis shows the class boundaries. The graph always starts at zero at the lowest class boundary and will end up at the total frequency (for a cumulative frequency) or 1.00 (for a relative cumulative frequency).

Example 6.1

You are working for the Transport manager of a large chain of supermarkets which hires cars for the use of its staff. He is interested in the weekly distances covered by these cars. Mileages recorded for a sample of hired vehicles from Fleet 1 during a given week yielded the following data:

138 164 150 132 144 125 149 157 161 150
 146 158 140 109 136 148 152 144 145 145
 168 126 138 186 163 109 154 165 135 156
 146 183 105 108 135 153 140 135 142 128

Minimum = 105 Maximum = 186 Range = 186 - 105 = 81

Nine intervals of 10 kilometres width is reasonable, but extreme intervals may be wider.

Next calculate the **frequencies** only within each interval.

Frequency density = frequency ÷ class width

<u>Class interval</u>	<u>Frequency</u>	<u>Frequency density</u>
100 and < 110	4	0.4
110 and < 120	0	
120 and < 130	3	0.3
130 and < 140	7	0.7
140 and < 150	11	0.11
150 and < 160	8	0.8
160 and < 170	5	0.5

170 and < 180	0	
180 and < 190	2	0.2
	40	

6.8.2 Simple Bar Charts

Bar Charts show the frequencies with which particular values of discrete variables, or particular attributes, occur, or the values attained by variables. In *Simple bar charts*, there is one bar for each value or attribute, and the heights of the bars are proportional to the frequencies or values attained as illustrated below. In *Multiple bar charts*, the heights of the bars are again proportional to the frequencies of the values attained, but bars are grouped together. In *compound bar charts*, each bar represents a total and is broken down into bands. The height of each band is proportional to the corresponding frequency or value attained. *Percentage bar charts* are like compound bar charts except that all bars are the same, and the heights of the bands show percentages of the total.

6.8.3 Cumulative Frequency Distribution

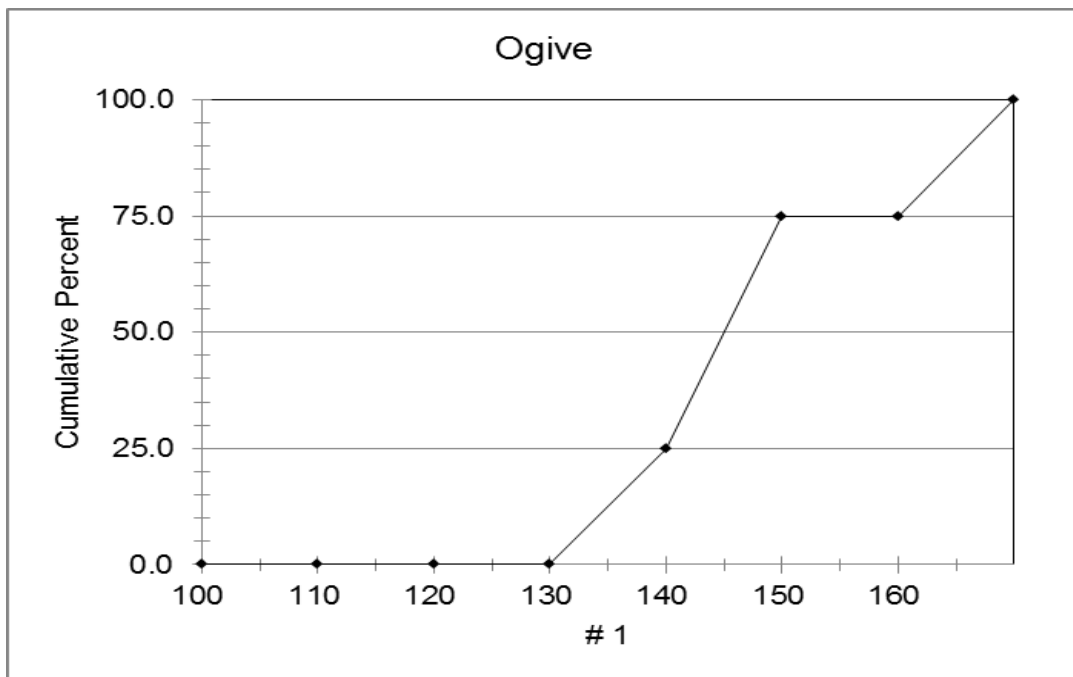
This has already been defined in the first section of this unit. Now, using data from Example 6.1 the cumulative frequency distribution, Ogive, and histogram are as follows:

Cumulative Frequency Distribution

<u>Interval less than</u>	<u>Frequency</u>	Cumulative <u>frequency</u>	% cumulative <u>frequency</u>
100	0	0	0.0
120	4	4	10.0
130	3	7	17.5
140	7	14	35.0

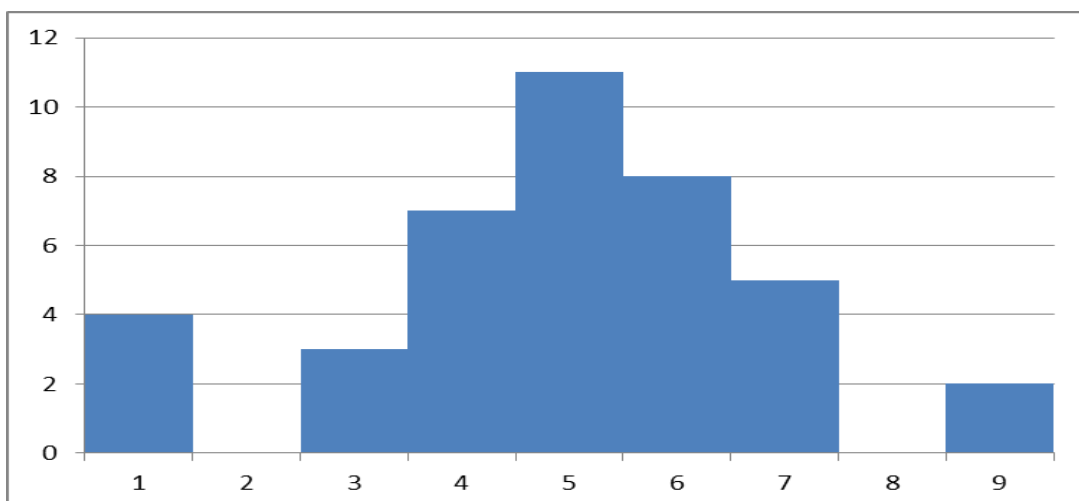
150	11	25	62.5
160	8	33	82.5
170	5	38	95.0
190	2	40	100.0

Figure 6.1 Ogive



Note: The graph is not a smooth curve because the data is discrete

Figure 6.2 Histogram



6.8.4 Measures of Central Tendency

An average is a measure of central tendency. The values of a variable may spread over a range, but these values cluster around a central point. This central point, or average, is in some way representative of the population as a whole. The most important averages are the arithmetic mean, the mode and the median.

Arithmetic Mean

This is the best known type of average.

For a set of ungrouped data, it is given by

$$\bar{x} = \frac{\sum x}{n} = \frac{\text{sum of data values}}{\text{number of data values}}$$

Example 6.2

Calculate the mean for the following set of data: 3, 5, 7, 12, and 15

Solution

$$\bar{x} = \frac{\sum x}{n} = \frac{\text{sum of data values}}{\text{number of data values}} = \frac{3+5+7+12+15}{5} = 8.4$$

For an ungrouped frequency distribution, it is given by

$$\bar{x} = \frac{\sum fx}{n} = \frac{\sum fx}{\sum f}$$

f = frequency of a class (=number of items in that class)

x = value of each item in that class

$n = \sum f$ = total number of items

Example 6.3

The number of jobs held by a group of young people since leaving school is as follows:

Number of jobs since leaving school	Frequency (number of people)
0	1
1	11
2	16
3	1
4	17
5	3
6	6

Calculate the mean number of jobs

Solution

x	F	fx
0	1	0
1	11	11
2	16	32
3	1	3
4	17	68
5	3	15
6	6	36

n=55	$\sum fx = 165$
------	-----------------

$$\bar{x} = \frac{\sum fx}{n} = \frac{165}{55} = 3$$

For a grouped frequency distribution, it is given by

$$\bar{x} = \frac{\sum fx}{\sum f}$$

f = frequency of a class (=number of items in that class)

x = mid-point of each range (class)

Example 6.4

The following frequency distribution is a record of monthly salaries for 50 employees in thousands of kwacha at a manufacturing plant:

Salary K'000	Number of employees
0 to less than 5	4
5 to less than 10	6
10 to less than 15	20
15 to less than 20	12
20 to less than 25	8

Calculate the mean salary

Solution

	Class	f	x	fx
	0 to less than 5	4	2.5	10
	5 to less than 10	6	7.5	45
	10 to less than 15	20	12.5	250
	15 to less than 20	12	17.5	210
$\bar{x} =$	20 to less than 25	8	22.5	180
		$\sum f = 50$		$\sum fx = 695$

$$\frac{\sum fx}{\sum f} = \frac{695}{50} = 13.9 \text{ that is K13,900}$$

Median

This is the midpoint of the data after being ranked (sorted in ascending order). There are as many numbers below the median as above the median.

For a set of ungrouped data, the median is the $\frac{1}{2}(n+1)^{th}$ term when the data is arranged from the smallest to the largest. n is the number of data values.

Example 6.5

Find the median for the following set of data; 20, 3, 9, 7, 13, 11, 24, 7, 20 and 27

Solution

Arranged in ascending order: 3, 7, 7, 9, 11, 13, 20, 20, and 27. $n=9$

$$\text{Therefore median} = \frac{1}{2}(9+1)^{th} \text{ term} = 5^{th} \text{ term} = 11$$

Find the median for the following set of data: 12, 13, 39, 15, 17, 9, 7, 20, 10, 29, 15, and 21

Solution

Arranged in ascending order: 7, 9, 10, 12, 13, 15, 15, 17, 20, 21, 29, and 39. $n=12$

Therefore median = $\frac{1}{2}(12+1)^{th} term = 6.5$ this implies it is between the 6th and 7th

$$term = \frac{15+15}{2} = 15$$

For a grouped frequency distribution the median can be estimated on an Ogive or calculated using the following formula;

$$\text{Median} = L + \frac{\left(\frac{N}{2} - F_{M-1}\right)C}{f_m}$$

L = lower limit of the median class. The median class is the class containing the $\frac{1}{2}(n+1)^{th}$ term

N = sum of the frequencies

F_{M-1} = cumulative frequency of the class immediately before the median class

f_m = actual frequency of the median class.

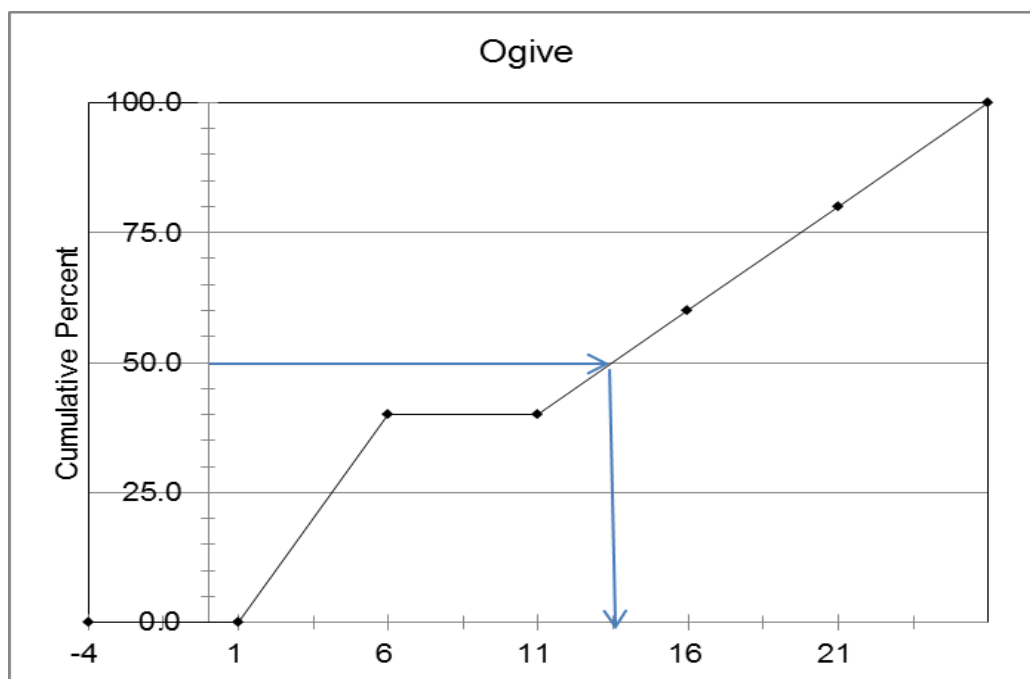
C = median class width

NOTE: the formula above for calculating the median is only applicable when there is a standard width (equal width).

Example 6.6

Estimate the median for the grouped frequency distribution in Example 6.4

Solution



The median is where the arrowed line cuts the horizontal axis at 13.5

Calculate the median for the grouped frequency distribution in Example 6.4

Class	f	Cf
0 to less than 5	4	4
5 to less than 10	6	10
10 to less than 15	20	30
15 to less than 20	12	42
20 to less than 25	8	50

Solution

$$\sum f = 50$$

$$\frac{1}{2} (50 + 1)^{th} \text{ term} =$$

25.5 that is between the 25th and 26th term. This is found in the class 10 to less than 15. Therefore;

$$L = 10$$

$$N = 50$$

$$F_{M-1} = 10$$

$$f_m = 20$$

$$C = 15 - 10 = 5$$

$$\text{Median} = 10 + \frac{\left(\frac{50}{2} - 10\right)5}{20} = 13.75$$

Mode

The mode is the most frequent data value in the distribution. For a set of ungrouped data, it is rather straightforward to find, but for a grouped frequency distribution, it is estimated on a histogram or calculated using the formula:

$$\text{Mode} = L + \left(\frac{D_1}{D_1 + D_2}\right)C$$

L = lower limit of the modal class. The modal class is the class with the highest frequency.

D_1 = difference between the highest frequency and the frequency of the class immediately before it.

D_2 = difference between the highest frequency and the frequency of the class immediately after it.

C = modal class width.

Example 6.7

Calculate the mode for the frequency distribution in Example 6.4

Modal class is 10 to less than 15. Therefore:

$$L = 10$$

$$D_1 = 20 - 6 = 14$$

$$D_2 = 20 - 12 = 6$$

$$C = 15 - 10 = 5$$

$$\text{Mode} = 10 + \left(\frac{14}{14 + 6} \right) 5 = 13.5$$

To end this sub-unit on measures of central tendency, here is a summary of some facts about the three measures, i.e. mean, median and mode.

Fact No.	The Mode	The Median	The Mean
1	It is the most frequent value in the distribution; it is the point of greatest density.	It is the value of the middle point of the array (not midpoint of range), such that half the items are above and half below it.	It is the value in a given aggregate which would obtain if all the values were equal.
2	The value of the mode is established by the predominant frequency, not by the value in the distribution.	The value of the median is fixed by its position in the array and doesn't reflect the individual value.	The sum of deviations on either side of the mean are equal; hence, the algebraic sum of the deviation is equal to zero.
3	It is the most probable value, hence the most typical.	The aggregate distance between the median point and all the values in the array is less than from any other point.	It reflects the magnitude of every value.
4	A distribution may have 2 or more modes (bimodal or polymodal). On the other hand, there is no mode in a rectangular distribution.	Each array has one and only one median, real or imaginary.	An array has one and only one mean, which can be a real number in the scores or imaginary.

5	The mode does not reflect the degree of modality.	It cannot be manipulated algebraically: medians of subgroups cannot be weighted and combined.	Means may be manipulated algebraically: means of subgroups may be combined when properly weighted.
6	It cannot be manipulated algebraically: modes of subgroups cannot be combined.	It is stable in that grouping procedures do not affect it appreciably.	It may be calculated even when individual values are unknown, provided the sum of the values and the sample size n are known.
7	It is unstable such that it is influenced by grouping procedures.	Value must be ordered, and may be grouped, for computation.	Values need not be ordered or grouped for this calculation.
8	Values must be ordered and grouped for its computation.	It can be computed when ends are open	It cannot be calculated from a frequency table when ends are open.
9	It can be calculated when table ends are open.	It is not applicable to qualitative data.	It is stable in that grouping procedures do not seriously affected (affect) it.
10	It is not affected by extreme values	It is not affected by extreme values	It is affected by extreme values

(Where a distribution is large and normal, all three measures of central tendency coincide and are the same on the normal or Gaussian curve)

6.8.5 Measures of Dispersion

Dispersion is the degree to which the data are spread out. Individual differences or variations exist. In describing a set of data, a measure of central tendency alone does not really tell us enough to make many decisions or inferences. Several distributions can have

the same mean yet the shape of the distribution of observations can be quite different. Thus to describe a set of data accurately, we typically use some measures of variation among the observations, along with a measure of central tendency. There are three measures of dispersion (central tendency) we will look at namely: the range, quartiles and the standard deviation.

Range

This is the difference between the highest and lowest values of a distribution. For a grouped frequency distribution, it is the difference between the upper limit of the highest class and the lower limit of the lowest class. It is not a very useful measure of dispersion hence nothing more than the definition will be outlined because it only gives an idea of the extreme values which might occur. It gives a vague idea of variability and spread.

Inter-quartile Range

The complete range of a set of data may be rather representative of the dispersion of a frequency distribution because there may be a few values far away from all the other values. Quartiles are a means of identifying the range over which most of the frequencies in a distribution occur.

The lower quartile, Q_1 is the value below which 25% of the distribution lies.

The upper quartile, Q_3 is the value below which 75% of the distribution lies.

For a set of ungrouped data, Q_1 is the $\frac{1}{4}(n+1)^{th}$ term and Q_3 is the $\frac{3}{4}(n+1)^{th}$ term where n is the number of data values and the data is arranged in ascending order and the answer in finding the term is rounded up to the next whole number.

For a grouped frequency distribution, quartiles can be estimated on an Ogive similar to the estimation of the median where the guide on the y axis is to use $\frac{1}{4} \sum f$ for Q_1 and $\frac{3}{4} \sum f$ for Q_3 . The inter-quartile range is given by $Q_3 - Q_1$. The semi-inter-quartile range is half the difference between the upper and lower quartile. The smaller the semi-inter-quartile range, the less spread out the distribution is.

Variance and Standard Deviation

The variance is average of the square deviations from the mean, the population variance, for ungrouped data, σ^2 , is given by;

$$\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{n}$$

The major disadvantage of the variance is that it gives squared units making it difficult to interpret, therefore, the standard deviation, σ is used which is the square root of the variance. It is given by;

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{n}} \quad \text{for ungrouped data}$$

The sample variance, s^2 and the sample standard deviation, for ungrouped data, s , are calculated using the following formulae

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} \quad \text{and} \quad s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

For a grouped frequency distribution, the variance and standard deviations are given by

$$\sigma^2 = \frac{\sum fx^2}{\sum f} - (\bar{x})^2 \quad \text{and} \quad \sigma = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2} \quad \text{for a population}$$

And;

$$s^2 = \frac{\sum fx^2 - \frac{(\sum fx)^2}{\sum f}}{\sum f - 1} \quad \text{and} \quad s = \sqrt{\frac{\sum fx^2 - \frac{(\sum fx)^2}{\sum f}}{\sum f - 1}} \quad \text{for a sample}$$

Example 6.8

Calculate the standard deviation for the following set of data; 2, 4, 7, 9 and 11

Solution

$$\bar{x} = \frac{2+4+7+9+11}{5} = 6.6$$

$$\sigma = \sqrt{\frac{(2-6.6)^2 + (4-6.6)^2 + (7-6.6)^2 + (9-6.6)^2 + (11-6.6)^2}{5}} = 3.262$$

Example 6.9

Calculate the standard deviation for the frequency distribution in example 3.4.1.3

Solution

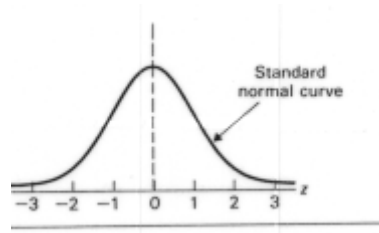
Class	f	x	fx	fx^2
0 to less than 5	4	2.5	10	25
5 to less than 10	6	7.5	45	337.5
10 to less than 15	20	12.5	250	3125
15 to less than 20	12	17.5	210	3675
20 to less than 25	8	22.5	180	4050
		$\sum f = 50$	$\sum fx = 695$	$\sum fx^2 = 112125$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{695}{50} = 13.9$$

$$\text{Standard deviation} = \sqrt{\frac{112125}{50} - 13.9^2} = \sqrt{31.04} = 5.57$$

6.9 INFERENCE STATISTICS

6.9.1 Normal Distribution



Any Normal Distribution has the following characteristics;

1. It is bell-shaped and is symmetrical about its mean.
2. It is asymptotic to the axis, i.e., it extends indefinitely in either direction from the mean to positive and negative infinity.
3. It is a continuous distribution (probability density function).
4. It is a family of curves, i.e., every unique pair of mean and standard deviation defines a different normal distribution. Thus, the normal distribution is completely described by two parameters: mean and standard deviation. See the following figure.
5. Total area under the curve sums to 1, i.e., the area of the distribution on each side of the mean is 0.5.
6. It is unimodal, i.e., values mound up only in the centre of the curve.
7. The probability that a random variable will have a value between any two points is equal to the area under the curve between those points. Any normal distribution can be transformed into a standard normal distribution such that the area or probability between any values of the distribution can be read off the standard normal tables. This is done using Z scores where:

$$Z = \frac{x - \mu}{\sigma}$$
 σ is the population standard deviation.

The standard normal table gives the probability between 0 and Z. A standard Z table can be used to find probabilities for any normal curve problem that has been converted into Z scores. There is a standard normal table in the appendix of the module. The Z distribution is a normal distribution with a mean of 0 and a standard deviation of 1. The following steps are helpful when working with the normal curve problems:

1. Graph the normal distribution, and shade the area related to the probability you want to

find.

2. Convert the boundaries of the shaded area from X values to the standard normal random variable Z values, using the Z formula above.
3. Use the standard Z table to find the probabilities or the areas related to the Z values in step 2.

Normal Probabilities

There is a table which must be used to look up standard normal probabilities. The z-score is broken into two parts, the whole number and tenth are looked up along the left side and the hundredth is looked up across the top. The value in the intersection of the row and column is the area under the curve between zero and the z-score looked up.

Because of the symmetry of the normal distribution, look up the absolute value of any z-score.

Computing Normal Probabilities

There are several different situations that can arise when asked to find normal probabilities.

Situation	Instructions
Between zero and any number	Look up the area in the table
Between two positives, or Between two negatives	Look up both areas in the table and subtract the smaller from the larger.
Between a negative and a positive	Look up both areas in the table and add them together
Less than a negative, or Greater than a positive	Look up the area in the table and subtract from 0.5000
Greater than a negative, or Less than a positive	Look up the area in the table and add to 0.5000

This can be shortened into two rules.

If there is only one z-score given, use 0.5000 for the second area, otherwise look up both z-scores in the table

If the two numbers are the same sign, then subtract; if they are different signs, then add. If there is only one z-score, then use the inequality to determine the second sign (< is negative, and > is positive).

Finding z-scores from probabilities

This is more difficult, and requires you to use the table inversely. You must look up the area between zero and the value on the inside part of the table, and then read the z-score from the outside. Finally, decide if the z-score should be positive or negative, based on whether it was on the left side or the right side of the mean. Remember, z-scores can be negative, but areas or probabilities cannot be.

Situation	Instructions
Area between 0 and a value	Look up the area in the table Make negative if on the left side
Area in one tail	Subtract the area from 0.5000 Look up the difference in the table Make negative if in the left tail
Area including one complete half (Less than a positive or greater than a negative)	Subtract 0.5000 from the area Look up the difference in the table Make negative if on the left side
Within z units of the mean	Divide the area by 2 Look up the quotient in the table Use both the positive and negative z-scores
Two tails with equal area	Subtract the area from 1.000 Divide the area by 2

(More than z units from the mean)	Look up the quotient in the table Use both the positive and negative z-scores
-----------------------------------	--

Using the table becomes proficient with practice, work lots of the normal probability problems!

Example 6.10

Packets of sweets are labelled as weighing 150 grams, but actual weights may be modelled by a normal distribution with mean 150 grams and standard deviation of 4

What percentage of packets will weigh less than 150 grams?

What percentage of packets will weigh more than 160 grams?

What percentage of packets will weigh less than 145 grams?

What percentage of packets will weigh more than 142 grams?

What percentage of packets will weigh less than 156 grams?

What percentage of packets will weigh between 152 and 158 grams?

What percentage of packets will weigh between 147 and 154 grams?

What percentage of packets will weigh between 140 and 146 grams?

Above how many grams will 10% of the packets lie?

Above how many grams will 86% of the packets lie?

Below how many grams will 40% of the packets of the sweets lie?

Solution

$$P(X < 150)$$

$$Z = \frac{150 - 150}{4} = 0$$

$$P(X > 150) = P(Z < 0) = 0.5 \text{ i.e. } 50\%$$

$$P(X > 160)$$

$$Z = \frac{160-150}{4} = 2.5$$

$$P(X>160) = P(Z>2.5) = 0.5 - P(0<Z<2.5) = 0.5 - 0.4938 = 0.0062 \text{ i.e. } 0.62\%$$

$$P(X<145)$$

$$Z = \frac{145-150}{4} = -1.25$$

$$P(X<145) = P(Z<-1.25) = 0.5 - P(-1.25<Z<0) = 0.5 - 0.3944 = 0.1056 \text{ i.e. } 10.56\%$$

$$P(X>142)$$

$$Z = \frac{142-150}{4} = -2$$

$$P(X>142) = P(Z>-2) = 0.5 + P(-2<Z<0) = 0.5 + 0.4772 = 0.9772 \text{ i.e. } 97.72\%$$

$$P(X<156)$$

$$Z = \frac{156-150}{4} = 1.5$$

$$\begin{aligned} P(X<156) &= P(Z<1.5) = 0.5 + P(0<Z<1.5) \\ &= 0.5 + 0.4332 = 0.9332 \text{ i.e. } 93.32\% \end{aligned}$$

$$P(152<X<156)$$

$$Z = \frac{152-150}{4} = 0.5 \qquad Z = \frac{156-150}{4} = 1.5$$

$$\begin{aligned} P(152<X<156) &= P(0.5<Z<1.5) \\ &= P(0<Z<1.5) - P(0<Z<0.5) \\ &= 0.4332 - 0.1915 = 0.2417 \text{ i.e. } 24.17\% \end{aligned}$$

$$P(147<X<154)$$

$$Z = \frac{147-150}{4} = -0.75 \qquad Z = \frac{154-150}{4} = 1$$

$$\begin{aligned}
P(147 < Z < 154) &= P(-0.75 < Z < 1) \\
&= P(-0.75 < z < 0) + P(0 < Z < 1) \\
&= 0.2734 + 0.3413 \\
&= 0.6147 \quad \text{i.e. 61.47\%}
\end{aligned}$$

$$P(140 < X < 146)$$

$$Z = \frac{140 - 150}{4} = -2.5 \qquad Z = \frac{146 - 150}{4} = -1$$

$$\begin{aligned}
P(140 < X < 146) &= P(-2.5 < Z < 0) - P(-1 < Z < 0) \\
&= 0.4938 - 0.3413 \\
&= 0.1525 \quad \text{i.e. 15.25\%}
\end{aligned}$$

Find Z such that the probability between 0 and Z is approximately 0.4. The closest is 0.3997 and its Z value is 1.28

$$\begin{aligned}
1.28 &= \frac{x - 150}{4} \\
x &= 150 + 1.28(4) = 155.12
\end{aligned}$$

Find Z such that the probability between 0 and Z is approximately 0.36. The closest is 0.3599 and its Z value is -1.08 since it is on the left.

$$\begin{aligned}
-1.08 &= \frac{x - 150}{4} \\
x &= 150 - 1.08(4) = 145.68
\end{aligned}$$

Find Z such that the probability between 0 and Z is approximately 0.1, the closest is 0.0987 and its Z value is -0.25 since it is on the left.

$$\begin{aligned}
-0.25 &= \frac{x - 150}{4} \\
x &= 150 - 0.25(4) = 149
\end{aligned}$$

6.9.2 Binomial Distribution

A binomial experiment is an experiment which satisfies these four conditions

A fixed number of trials.

Each trial is independent of the others.

There are only two outcomes, a desired outcome (success) and undesired outcome (failure).

The probability of each outcome remains constant from trial to trial.

These can be summarized as: An experiment with a fixed number of independent trials, each of which can only have two possible outcomes.

A binomial or Bernoulli trial experiment has a *fixed number of independent trials*, each with only *two outcomes, yes or no, success or failure, among others*.

The fact that each trial is independent actually means that the probabilities remain constant.

Examples of binomial experiments

Tossing a coin 20 times to see how many tails occur.

Asking 200 people if they watch ZNBC news.

Rolling a die to see if a 5 appears.

Asking 500 die-hard MMD cadres, if they would vote for UPND. (Just because something is unlikely, doesn't mean that it isn't binomial. The conditions are met - there's a fixed number [500], the trials are independent [what one person does doesn't affect the next person], and there are only two outcomes [yes or no]).

Examples which aren't binomial experiments

Rolling a die until a 6 appears (not a fixed number of trials)

Asking 20 people how old they are (not two outcomes)

Drawing 5 cards from a deck for a poker hand (done without replacement, so not independent)

Binomial Probability Function

Example 6.11

What is the probability of rolling exactly two sixes in 6 rolls of a die?

Solution

There are five things you need to do to work out a binomial story problem.

Define Success first. Success must be for a single trial. Success = "Rolling a 6 on a single die"

Define the probability of success (p): $p = 1/6$

Find the probability of failure: $q = 5/6$

Define the number of trials: $n = 6$

Define the number of successes out of those trials: $x = 2$

Anytime a six appears, it is a success (denoted S) and anytime something else appears, it is a failure (denoted F). The ways you can get exactly 2 successes in 6 trials are given below. The probability of each is written to the right of the way it could occur. Because the trials are independent, the probability of the event (all six dice) is the product of each probability of each outcome (die)

$$1 \text{ FFFFSS } \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$2 \text{ FFFSFS } \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$3 \text{ FFFSSF } \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$4 \text{ FFSFSS } \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$5 \text{ FFSFSF } \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$6 \text{ FFSSFF } \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$7 \text{ FSFFFF } \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$8 \text{ FSFFSF } \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$9 \text{ FSFSFF } \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$10 \text{ FSSFFF } \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$11 \text{ SFFFFS} \quad \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$12 \text{ SFFFSF} \quad \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$13 \text{ SFFSFF} \quad \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$14 \text{ SFSFFF} \quad \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

$$15 \text{ SSFFFF} \quad \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$$

Notice that each of the 15 probabilities are exactly the same: $\left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4$

Also, note that the $\frac{1}{6}$ is the probability of success and you needed 2 successes. The $\frac{5}{6}$ is the probability of failure, and if 2 of the 6 trials were success, then 4 of the 6 must be failures. Note that 2 is the value of x and 4 is the value of $n-x$.

Further note that there are fifteen ways this can occur. This is the number of ways 2 successes can occur in 6 trials without repetition and order not being important, or a combination of 6 things, 2 at a time.

The probability of getting exactly x success in n trials, with the probability of success on a single trial being p is:

$$P(X=x) = {}^nC_x \times p^x \times q^{(n-x)}$$

Example 6.12

The pass rate for BAccEd candidates is 60%. Out of 8 BAccEd candidates, find the probability that:

(i) At least two candidates pass.

(ii) Six or Seven candidates pass.

Solution

$$p = 0.6 \quad q = 0.4 \quad n = 8 \quad P(X = x) = {}^8C_x (0.6)^x (0.4)^{8-x}$$

$$\begin{aligned}
P(\text{at least } 2) &= P(X \geq 2) = 1 - P(X < 2) \\
&= 1 - [P(X = 0) + P(X = 1)] \\
&= 1 - [8C_0(0.6)^0(0.4)^8 + 8C_1(0.6)^1(0.4)^7] \\
&= 1 - (0.00065536 + 0.00786432) \\
&= 1 - (0.00851968) \\
&= 0.99148032
\end{aligned}$$

$$\begin{aligned}
P(X = 6 \text{ or } 7) &= P(X = 6) + P(X = 7) \\
&= 8C_6(0.6)^6(0.4)^2 + 8C_7(0.6)^7(0.4)^1 \\
&= 0.20901888 + 0.08957952 \\
&= 0.2985984
\end{aligned}$$

6.9.3 Mean, Variance, and Standard Deviation

The mean, variance, and standard deviation of a binomial distribution are extremely easy to find.

$ \begin{aligned} \mu &= np \\ \sigma^2 &= npq \\ \sigma &= \sqrt{npq} \end{aligned} $
--

6.9.4 Poisson Distribution

Named after the French mathematician Simeon Poisson, Poisson probabilities are useful when there are a large number of independent trials with a small probability of success on a single trial, and the variables occur over a period of time. It can also be used when a density of items is distributed over a given area or volume. It is used for random and rare events which have an average occurrence. It can also be used as an approximation to the binomial when n is large, and p very small.

$$P(X = x) = \frac{e^{-m} m^x}{x!}$$

Where m represents the mean and the standard deviation is $\sqrt{m}$

For example,

- The number of phone calls received by a telephone operator in a 10-minute period.
- The number of patients arriving at a hospital per hour.
- The number of typing errors per page made by a secretary.

Example 6.13

On a production line, the probability of finding a defective product is 0.002. In a batch of 500 products, find the probability of finding exactly 2 defective products?

Solution

$$m = np = 500 \times 0.002 = 1$$

$$\text{therefore } P(X = x) = \frac{e^{-1} 1^x}{x!} \text{ now } P(X = 2) = \frac{e^{-1} 1^2}{2!} = 0.183939721$$

$$\text{using the binomial } P(X = x) = {}^{500}C_x (0.002)^x (0.998)^{500-x}$$

$$\text{Now } P(X = 2) = {}^{500}C_2 (0.002)^2 (0.998)^{498} = 0.184123875$$

Example 6.14

The average number of emergency patient arrivals at a hospital every 30 minutes is 2. Find the probability that:

- Over a 30 minutes period no patient arrives.
- Over a 30 minutes period exactly two patients arrive.
- During an hour, at least two patients arrive.
- During a two hour period less than 2 patients arrive.

Solution

$$\text{a) } m = 2 \quad P(X = x) = \frac{e^{-2} 2^x}{x!}$$

$$P(X = 0) = \frac{e^{-2} 2^0}{0!} = e^{-2} = 0.135335283$$

b) $m = 2 \quad P(X = x) = \frac{e^{-2} 2^x}{x!}$

$$P(X = 2) = \frac{e^{-2} 2^2}{2!} = \frac{e^{-2} 4}{2} = 2e^{-2} = 0.270670568$$

c) 30 min = 2 patients

Therefore in 60 min (1hr) = 4 patients and $m = 4 \quad P(X = x) = \frac{e^{-4} 4^x}{x!}$

$$P(\text{at least } 2) = P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - [P(x=0) + P(X=1)]$$

$$= 1 - \left[\frac{e^{-4} 4^0}{0!} + \frac{e^{-4} 4^1}{1!} \right]$$

$$= 1 - [e^{-4} + 4e^{-4}]$$

$$= 1 - 5e^{-4} = 1 - 0.091578194 = 0.908421806$$

d) 30 min = 2 patients

Therefore for 8 patients $m = 8 \quad P(X = x) = \frac{e^{-8} 8^x}{x!}$

$$P(\text{less than } 2) = P(X < 2) = P(X=0) + P(X=1)$$

$$= \frac{e^{-8} 8^0}{0!} + \frac{e^{-8} 8^1}{1!}$$

$$= e^{-8} + 8e^{-8} = 9e^{-8} = 0.003019164$$

6.9.5 Methods of Sampling and Survey Design

Sampling techniques

These are methods by which the researcher can derive a sample from a population. Naturally, if the aim of a certain study is to learn things about a certain population, the optimum methodology is to test all members of that population. However, as a rule, you simply cannot do this, since it can be prohibitively time-consuming and expensive, and ultimately pointless.

Random Sampling

Random sampling is a sampling technique where we select a group of subjects (a sample) for study from a larger group (a parent population). Each individual is chosen entirely by chance and each member of the population has a known, but possibly non-equal, chance of being included in the sample.

By using random sampling, the likelihood of bias is reduced.

Simple Random Sampling

Simple random sampling is the basic sampling technique where we select a group of subjects (a sample) for study from a larger group (a population). Each individual is chosen entirely by chance and each member of the population has an equal chance of being included in the sample. Every possible sample of a given size has the same chance of selection; that is, each member of the population is equally likely to be chosen at any stage in the sampling process.

Stratified Sampling

There may often be factors which divide up the population into sub-populations (groups / strata) and we may expect the measurement of interest to vary among the different sub-populations. This has to be accounted for when we select a sample from the population in order that we obtain a sample that is representative of the population. This is achieved by stratified sampling.

A stratified sample is obtained by taking samples from each stratum or sub-group of a population.

When we sample a population with several strata, we generally require that the proportion of each stratum in the sample should be the same as in the population.

Stratified sampling techniques are generally used when the population is heterogeneous, or dissimilar, where certain homogeneous, or similar, sub-populations can be isolated (strata). Simple random sampling is most appropriate when the entire population from which the sample is taken is homogeneous. Some reasons for using stratified sampling over simple random sampling are:

- a) The cost per observation in the survey may be reduced;
- b) Estimates of the population parameters may be weighted for each sub-population;
- c) Increased accuracy at given cost.

Types of Parameter estimates

These two types of **estimate** of a **population parameter** are referred to as:

- Point estimate - one particular value;
- Interval estimate - an interval centred on the point estimate.

A good estimator must satisfy three conditions:

- Unbiased: The expected value of the estimator must be equal to the mean of the parameter
- Consistent: The value of the estimator approaches the value of the parameter as the sample size increases
- Relatively Efficient: The estimator has the smallest variance of all estimators which could be used

Point Estimates of Population Parameters

From the sample, a value is calculated which serves as a point estimate for the population parameter of interest.

- a) The best estimate of the population **percentage**, π , is the sample percentage, p .
- b) The best estimate of the unknown population **mean**, μ , is the sample mean, $\bar{x} = \frac{\sum x}{n}$.

This estimate of μ is often written $\hat{\mu}$ and referred to as 'mu hat'.

c) The best estimate of the unknown population **standard deviation**, σ , is the sample standard deviation s , where:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{(n-1)}} \quad \text{This is obtained from the } x\sigma_{n-1} \text{ key on the calculator.}$$

N.B. $s = \sqrt{\frac{\sum (x - \bar{x})^2}{(n)}}$ from $x\sigma_n$ is **not** used as it underestimates the value of σ .

Sampling Distribution of the Sample Means

Instead of working with individual scores, statisticians often work with means. What happens is that several samples are taken, the mean is computed for each sample, and then the mean of sample means is used as the data, rather than individual scores being used. The sample is a sampling distribution of the sample means.

Properties of the Sampling Distribution of the Sample Means

When all of the possible sample means are computed, then the following properties are true:

The mean of the sample means will be the mean of the population

The variance of the sample means will be the variance of the population divided by the sample size.

The standard deviation of the sample means (known as the standard error of the mean) will be smaller than the population mean and will be equal to the standard deviation of the population divided by the square root of the sample size.

If the population has a normal distribution, then the sample means will have a normal distribution.

If the population is not normally distributed, but the sample size is sufficiently large, then the sample means will have an approximately normal distribution. Some books define sufficiently large as at least 30 and others as at least 31. This is known as the central limit theorem. The theorem states that as the sample size gets large, and different samples are taken from the same parent population of interest, it will tend to have the sample statistic or mean of the sample means approaching the population mean or parameter.

The formula for a z-score when working with the sample means is:
$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

6.9.6 Confidence intervals

Sometimes it is more useful to quote two limits between which the parameter is expected to lie, together with the probability of it lying in that range.

The limits are called the **confidence limits** and the interval between them the **confidence interval**.

The width of the confidence interval depends on three sensible factors:

- a) The degree of confidence we wish to have in it, i.e. the probability of it including the 'truth', e.g. 95%;
- b) The size of the sample, n;
- c) The amount of variation among the members of the sample, e.g. for means this is the standard deviation, s.

The **confidence interval** is therefore an interval centred on the point estimate, in this case either a percentage or a mean, within which we expect the population parameter to lie. The width of the interval is dependent on the confidence we need to have; it does in fact include the population parameter, the size of the sample, n, and its standard deviation, s, if estimating means. These last two parameters are used to calculate the **standard error**, $s/\sqrt{n}$, which is also referred to as the standard deviation of the mean.

The number of standard errors included in the interval is found from statistical tables - either the normal or the t-table. Always use the normal tables for percentages which need large samples. For means, the choice of table depends on the sample size and the population standard deviation:

Distribution	Sample size	Population standard deviation	
		Known: standard error = $\frac{\sigma}{\sqrt{n}}$	Unknown: standard error = $\frac{s}{\sqrt{n}}$
Normal	Large	Normal tables	Normal tables
Normal	Small	Normal tables	t-tables
Non-normal	large	Normal tables	Normal tables
Non-normal	small	Not available	Not available

Interpretation of Confidence intervals

How do we interpret a confidence interval? If 100 similar samples were taken and analysed then, for a 95% confidence interval, we are confident that 95 of the intervals calculated would include the true population mean. In practice we tend to say that we are 95% confident that our interval includes the true population value. Note that there is only one true value for the population mean. It is the variation between samples which gives the range of confidence intervals.

If the population has a normal distribution with known variance, a confidence interval for the population mean can be calculated as;

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \text{ where,}$$

$z_{\frac{\alpha}{2}}$ = reliability factor, a standard normal random variable for which the probability in the right-hand tail of the distribution is $\frac{\alpha}{2}$. In other words, this is the z-score that leaves $\frac{\alpha}{2}$ in the upper tail.

Confidence Interval and Hypothesis Testing for a Single Mean

The methods of interval estimation will be based on the assumption that the normal probability distribution can be used. Such use is warranted whenever, n, the sample size is

at least 30 because of the central limit theorem or when n is less than 30 but the population is normally distributed and σ is known.

A confidence interval for the mean is an estimate interval with respect to the sample mean by which the likelihood that the interval includes the value of the population mean can be specified. The level of confidence associated with a confidence interval indicates the long run percentage of such intervals which would include the parameter being estimated. In other words, a confidence interval is an interval estimate with a specific level of confidence where a confidence level is the percentage of the time the true mean will lie in the interval estimate given.

If $\bar{x}$ is the mean of a random sample of size n taken from a normal population with standard deviation σ , then the confidence interval for μ , the population mean is given by,

$$\bar{x} \pm Z_c \frac{\sigma}{\sqrt{n}}$$

Where Z_c is the critical value of Z from the normal tables. $\pm Z_c \frac{\sigma}{\sqrt{n}}$ is called the margin of error of the confidence interval and it is one half of the width of the confidence interval. The most frequently used confidence intervals are 95%, 98%, and 99%.

The 95% confidence interval is given by $\bar{x} \pm 1.96 \frac{s}{\sqrt{n}}$

The 98% confidence interval is given by $\bar{x} \pm 2.33 \frac{s}{\sqrt{n}}$

The 99% confidence interval is given by $\bar{x} \pm 2.58 \frac{s}{\sqrt{n}}$

When σ is not known we use the sample standard deviation s

Example 6.15

The distances travelled to work (in km) by a sample of 30 employees of a company are as follows:

9 2 15 5 8 10

10 20 5 15 9 8

10 15 5 10 8 1

10 10 9 12 7 10

11 10 20 25 10 7

Calculate the 95% interval for the mean and interpret the result.

Solution

$$\bar{X} = \frac{9(3) + 10(9) + 11 + 2 + 20(2) + 15(3) + 5(3) + 12 + 25 + 8(3) + 1 + 7(2)}{30} = 10.2$$

$$s = \sqrt{\frac{3(9 - 10.2)^2 + 9(10 - 10.2)^2 + (11 - 10.2)^2 + 2(20 - 10.2)^2 + 3(15 - 10.2)^2 + 3(5 - 10.2)^2 + (12 - 10.2)^2 + (25 - 10.2)^2 + 3(8 - 10.2)^2 + (1 - 10.2)^2 + 2(7 - 10.2)^2}{30 - 1}} \\ = 5.108$$

Critical value from the normal tables for 95% confidence interval is 1.96

Therefore the 95% confidence interval for the population mean is;

$$(10.2 \pm 1.96 \frac{5.108}{\sqrt{30}}) = (10.2 \pm 1.828)$$

$$= (10.2 - 1.828) \text{ to } (10.2 + 1.828)$$

$$= 8.372 \text{ to } 12.028$$

This implies that with a probability of 95%, the average distance travelled by all employees is between 8.372 and 12.028 kilometres.

The Student's t Test

The t distributions were discovered in 1908 by William Gosset, who was a chemist and a statistician employed by the Guinness brewing company. He considered himself a student still learning statistics, so that is how he signed his papers as pseudonym "Student". The shape of a t-distribution depends on a parameter called "degree-of-freedom". As the degree-of-freedom gets larger, the t-distribution gets closer and closer to the standard

normal distribution. Recall that in estimating the population's variance, we used (n-1) rather than n, in the denominator. The factor (n-1) is called "degrees of freedom."

For practical purposes, the t-distribution is treated as the standard normal distribution when degree-of-freedom is greater than 30. Therefore, we use it for estimation when the sample size is less than 30. The area under the t distribution is found using the t tables

If the distribution of the population is normal with unknown variance, we can use the t-distribution to construct a confidence interval:

$\bar{X} \pm t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$ where $t_{\frac{\alpha}{2}}$ = reliability factor corresponding to a t-distribution random variable with $n - 1$ degrees of freedom, where n is the sample size. The area under the tail of the t-distribution to the right of $t_{\frac{\alpha}{2}}$ is $\frac{\alpha}{2}$.

Example 6.16

A large firm randomly monitored its own effluent before releasing it to the treatment plant. The concentration, in parts per million, of a certain chemical were found to be:

5.8 6.0 6.1 5.8 5.2 5.7 6.8 5.3 5.6 6.4

Find a 95% confidence interval for the concentration of the chemical in the effluent.

Solution

$$\bar{X} = \frac{5.8(2) + 6 + 6.1 + 5.2 + 5.7 + 6.8 + 5.3 + 5.6 + 6.4}{10} = 5.87$$

$$s = \sqrt{\frac{2(5.8 - 5.87)^2 + (6 - 5.87)^2 + (6.1 - 5.87)^2 + (5.2 - 5.87)^2 + (5.7 - 5.87)^2 + (6.8 - 5.87)^2 + (5.3 - 5.87)^2 + (5.6 - 5.87)^2 + (6.4 - 5.87)^2}{10 - 1}} = 0.483$$

Degrees of freedom = 10 - 1, critical value from the t table is 1.833

$$95\% \text{ confidence interval} = (5.87 \pm 1.833 \frac{0.483}{\sqrt{10}}) = (5.87 \pm 0.28) = (5.87 - 0.28) \text{ to } (5.87 + 0.28) = 5.59 \text{ to } 6.15$$

Hypothesis Testing for a Single Mean

In section 4.15.2, we estimated an unknown population parameter from the corresponding statistic obtained from the analysis of a sample from the population. In this section, we shall similarly analyse a sample and then use its statistic to see whether some claim made about the population is reasonable or not.

When an estimate from a **sample** is used to test some belief, claim or **hypothesis** about the actual **population**, the process is known as **hypothesis testing** (Significance Testing).

Some **conclusion** about the **population** is drawn from evidence provided by the sample. A formal procedure is used for this technique.

The Method which is common to all Hypothesis Tests:

(E.g. A claim has been made that average male salary in a particular firm is K15 000)

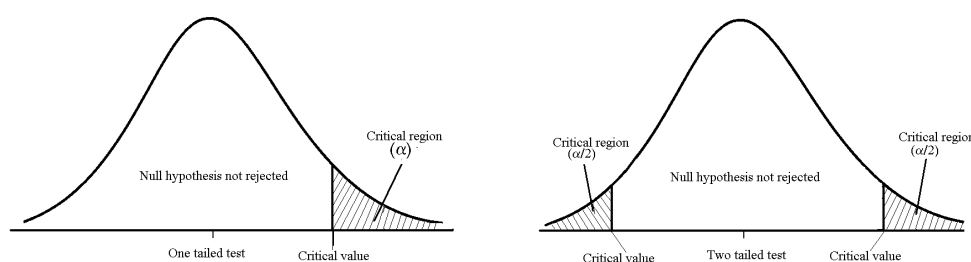
1. State the null hypothesis (H_0) - This is a statement about the population which may, or may not, be true. It takes the form of an equation making a claim about the population. $H_0: \mu = \text{K15 000}$ where μ is the population mean.
2. State the alternative hypothesis (H_1) - the conclusion reached if the Null Hypothesis is rejected. It will include the terms: not equal to ($\neq$), greater than ($>$), or less than ($<$).
 - a. $H_1: \mu \neq \text{K15 000}$, **or** $H_1: \mu < \text{K15 000}$, **or** $H_1: \mu > \text{K15 000}$
3. State the significance level (α) of the test - the proportion of the time you are willing to reject H_0 when it is in fact true. If not stated specifically, a 5% (0.05) significance level is used.
4. Find the critical value - this is the value from the appropriate table which the test statistic is required to reach before we can reject H_0 . It depends on the significance level, α , and whether the test is one- or two-tailed.
5. Calculate the test statistic - This value is calculated from the data obtained from the sample. From the data we find the sample statistics, such as the mean and standard deviation, which we need for calculating the test statistic; these are then substituted

into the appropriate formula to find its value. (Each formula is used in one specific type of hypothesis test only and so it needs careful selection.).

6. Reach a conclusion - Compare the values of the Test Statistic and the Critical Value. If the Test Statistic is greater than the Critical Value then we have enough evidence from the sample data to cause us to reject the null hypothesis and conclude that the Alternative Hypothesis is true, e.g. the mean salary is not K15 000 . If the Test Statistic is not greater than the Critical Value, then we do not have sufficient evidence and so do not reject the null hypothesis, e.g. the mean salary for the whole firm could be K15 000 .

(NB We can prove that the Null Hypothesis is false but we can never prove it to be true since we do not have all the evidence possible as only a sample is available for analysis.)

Graphical interpretation of conclusion



If the test statistic falls into the critical region, there is sufficient evidence to cause the null hypothesis to be rejected. The total shaded area is the significance level, α (alpha), as either a percentage (5%) or a decimal (0.05).

As illustrated above, there are two types of tests, one tailed and two tailed tests. A one tailed test looks for a definite increase or decrease in the population parameter while a two tailed test looks for a difference in the parameter

TESTING FOR THE MEAN OF A POPULATION

The methods described in this section require the population to be measured on the interval or ratio scales and to be normally distributed. You may assume this to be correct for any data presented here.

As with confidence intervals, before deciding which test to use, we have to ask whether the sample size is large or small and whether the population standard deviation is known or has to be estimated from the sample. The methods used are basically the same but different formulae and tables need to be used.

Summarising again the choice of statistical table:

Sample size	Population standard deviation	
	Known: standard error = $\frac{\sigma}{\sqrt{n}}$	Unknown: standard error = $\frac{s}{\sqrt{n}}$
Large	Normal tables: z-test	Normal tables: z-test
Small	Normal tables: z-test	t-tables: t-test

Method

A **mean value** is hypothesised for the population. A sample is taken from that population and its mean value calculated. This sample mean is then used to see if the value hypothesised for the population is reasonable or not. If the population standard deviation is not known, then that from the sample is calculated and used to estimate it. The appropriate test to use, either a t-test or a z-test, depends on whether this estimated value has been used or not.

H₀: $\mu = c$ **H₁:** $\mu \neq c$, $\mu < c$, or $\mu > c$, where μ is the population mean and c is the hypothesised value

Significance Level: $\alpha = 5\%$, or as stated in the question.

Critical value From normal (z) table or t-tables, significance level, number of tails, degrees of freedom.

If we know the population standard deviation and do not need to estimate it, the normal table is used. If we need to estimate it, then a slightly larger critical value is found from the t-table.

Test statistic

$$z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \quad \text{or} \quad t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

where $\bar{x}$ and μ are the means of the sample and population respectively and s and σ are their standard deviations.

Conclusion Compare the test statistic with the critical value. Decide whether to reject H_0 or not. Conclude in terms of question.

Example 6.17

A survey of all local employees last year indicated that the average hours worked per worker per week was 39 hours. A company has taken a random sample of 49 employees, and has found that the average number of hours worked per week was 40 hours, with a variance of 9 hours².

Determine whether the average number of hours worked per week by the company's employees is significantly different from the local average for last year at 5% level of significance.

Solution

$$H_0: \mu = 39$$

$$H_1: \mu \neq 39$$

We perform a two-tailed test with critical values of Z as ± 1.96

$$\text{Test statistic } Z = \frac{40 - 39}{\frac{3}{\sqrt{49}}} = 2.33$$

Conclusion: since the test statistic lies outside the critical value range, the null hypothesis is rejected. Therefore, there is a significant difference between the average number of hours worked per week by the company's employees and the local average for last year.

Example 6.18

A paint manufacturer claims that one tin of paint will cover an average of at least 80 square meters with a standard deviation of 10 square meters. A government inspector wants to test the validity of this claim. He takes a sample of 36 tins and finds that the average is 75 square meters. Test the manufacturer's claim at 5% significance level.

Solution

$$H_0 : \mu \geq 80$$

$$H_1 : \mu < 80$$

$$\text{Test statistic} = \frac{75 - 80}{\sqrt{\frac{10}{36}}} = -3$$

Critical value from the standard normal tables is $Z = -1.65$ since it is a one tailed test.

Since the test statistic is less than the critical value, H_0 is rejected. Therefore, the manufacturer's claim is not acceptable.

Example 6.19

The average weight of metal rod is expected to be greater than 79 grams with standard deviation 7 grams. A sample of 25 rods produced an average weight of 85 grams. Test the claim at 1% level of significance.

Solution

$$H_0 : \mu \geq 79$$

$$H_1 : \mu < 79$$

Since the sample size is small we use the t distribution with $df = 25 - 1 = 24$

$$\text{Test statistic } t = \frac{85 - 79}{\frac{7}{\sqrt{25}}} = -4.29$$

Critical value from the t distribution is -2.492 since it is a one tailed test.

Since the test statistic is less than the critical value, the null hypothesis is rejected.

Therefore there is sufficient evidence at 1% level to reject the claim

Confidence Interval and Hypothesis Testing for a Difference Between Two Means

Confidence Intervals for the difference between two means

When two independent samples are drawn from two independent normal distributions, the difference between the means is known as the sampling distribution of the difference between means. There is often need to estimate the difference between two population means, such as the difference between wage levels in two firms.

The formula used for estimating the difference between two population means with confidence intervals is,

$$\bar{X}_1 - \bar{X}_2 \pm Z_c \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad \text{when the standard deviations of the two populations are known}$$

Or,

$$\bar{X}_1 - \bar{X}_2 \pm Z_c \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad \text{when the standard deviations of the two populations are not known}$$

Note that the formulae above are used when both samples are greater than or equal to 30.

When both samples are less than 30, the confidence interval for the difference between two means is given by:

$$\bar{X}_1 - \bar{X}_2 \pm t_c \sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}$$

Where $s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$ with $df = n_1 + n_2 - 2$, s_p^2 is called the pooled variance.

It is an estimate of the common population variance for the two samples

Example 6.20

A random sample of 50 households in community A has a mean household income of K4,460 with a standard deviation K220. A random sample of 50 households in community B has a mean of K4,380 with a standard deviation of K280. Estimate the difference in the average household income in the two communities using a 95% confidence interval

Solution

$$\begin{aligned} 95\% \text{ confidence interval} &= (4460 - 4380) \pm 1.96 \sqrt{\frac{220^2}{50} + \frac{280^2}{50}} \\ &= 80 \pm 98.7 \\ &= (80 - 98.7) \text{ to } (80 + 98.7) \\ &= -18.7 \text{ to } 178.7 \end{aligned}$$

Example 6.21

The mean weekly wage for a sample of 12 employees in a large manufacturing firm is K280 with a sample standard deviation of K14. In another large firm a random sample of 10 employees has a mean weekly wage of K270 with a sample standard deviation of K10. Calculate a 99% confidence interval for the difference between the mean weekly wage levels in the two firms.

Solution

Since the sample sizes are small we get the pooled variance and use the t distribution with $df = 12 + 10 - 2 = 20$.

$$s_p^2 = \frac{(12-1)14^2 + (10-1)10^2}{12+10-2} = 152.8$$

$$\begin{aligned} 99\% \text{ Confidence interval} &= (280 - 270) \pm 2.845 \sqrt{\frac{152.8}{12} + \frac{152.8}{10}} \\ &= 10 \pm 1.218 \\ &= (10 - 1.218) \text{ to } (10 + 1.218) \end{aligned}$$

$$= 8.782 \text{ to } 11.218$$

Hypothesis Testing for Difference Between two Means

The population parameters which are of interest in this section are the population means, μ_1 and μ_2 . To perform inference about the difference between two means, we must first check if the two samples on hand had been taken as independent or dependent samples. A sampling method is independent when the individuals in one sample do not dictate which individuals are to be taken for the second sample. A sampling method is dependent when the individuals selected to represent the first population are used to determine the individuals to be included in the second sample that is representing the second population. The interest, in this section, lies in the difference between those two means for the two populations when the samples have been taken independently.

Classical Method Steps: when population variances are known, large samples

1. State the null hypothesis. H_0
2. Decide on the alternative hypothesis, H_1
 - a) Equal hypothesis versus not equal hypothesis: $H_0 : \mu_1 = \mu_2$ versus $H_1 : \mu_1 \neq \mu_2$, two tailed test.
 - b) At least versus less than: $H_0 : \mu_1 \geq \mu_2$ versus $H_1 : \mu_1 < \mu_2$, left-tailed one tailed test.
 - c) At most versus greater than: $H_0 : \mu_1 \leq \mu_2$ versus $H_0 : \mu_1 > \mu_2$, right-tailed one tailed test.
3. Choose the level of significance, thus fixing the critical region.
Let α be the significance level. The three cases that will go along to find the critical values and the rejection regions for the test will be found as in section 4.15
4. Calculate the standardised z statistic.

The test statistic we have $Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$, where n_1 and n_2 are samples

sizes for the two samples, $\bar{x}_1$ and $\bar{x}_2$ are the sample means and Z has the standard normal distribution.

The above test statistic is computed based on the information provided to us by the sample data.

The statistical decision made will be based on the case on hand, whether we have a two-tailed, a left-tailed or a right-tailed test, by comparing the computed value of the test statistic to the critical value, based on the test being chosen.

5. Accept or reject the null hypothesis

Classical Method Steps: when population variances are unknown, large samples.

In this situation a sample size is considered large if $n \geq 30$. Populations can be assumed to be normally distributed with the population means, μ_1 and μ_2 , and populations' variances σ_1^2 and σ_2^2 .

The steps will go as it was when the population variances are known with the exception that the test statistic will take a different form. It is still a 2-sample z-test with the sample variances replacing the populations' variances in the form for the test statistic. Hence the test statistic will have the following form:

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Classical Method Steps: when population variances are unknown, Small Samples

The test statistic will be given by

$$T = Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{Where } s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2} \quad \text{with df} = n_1 + n_2 - 2,$$

s_p^2 is called the pooled variance. It is an estimate of the common population variance for the two samples.

Example 6.22

The average annual membership fee at a sample of 200 sports clubs in the south-west region of a country is K250 with a standard deviation of K45. The average annual membership fee at a sample of 150 sports clubs in the north-east region is K220 with a standard deviation of K55. Using a 5% significance level, test the null hypothesis that average sports club membership fees are the same in both regions.

Solution

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

The critical value region from the normal tables is $Z = \pm 1.96$.

$$\text{Test statistic } Z = \frac{250 - 220}{\sqrt{\frac{45^2}{200} + \frac{55^2}{150}}} = 5.45$$

Since the test statistic lies outside the critical value range, H_0 is rejected. Therefore, there is no difference in the average sports club membership fees in both regions.

Example 6.23

A bank has two regional head offices, in towns B and L. Bank workers in the L office claim that they are paid less than the workers in the B office. Random samples taken from the two offices yield the following results:

	B	L
Sample size:	100	100
Mean salary:	K2790	K2625
Standard deviation:	K2,000	K2,100

You are asked to test the claim made by the L workers at 2% level of significance

Solution

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

The critical value is $Z = -2.05$ since this is a left sided one-tailed test.

$$\text{Test statistic } Z = \frac{2790 - 2625}{\sqrt{\frac{2000^2}{100} + \frac{2100^2}{100}}} = 0.57$$

Since the test statistic is greater than the critical value, H_0 is not rejected. There is sufficient evidence to justify the claim by the workers from the L office.

Example 6.24

The average annual expenditure on home insurance in a sample of 15 households in the south of a country is K180 with a standard deviation of K50. The average annual expenditure on home insurance in a sample of 14 households in the north of the country is K175 with a standard deviation of K60. Using a 1% significance level, test the null hypothesis that the average expenditure on home insurance is the same in both regions.

Solution

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Since the sample sizes are small $df = 15 + 14 - 2 = 27$, hence the critical value region from the t tables is from -2.771 to +2.771

$$s_p^2 = \frac{(15-1)50^2 + (14-1)60^2}{15+14-2} = 55.042$$

$$\text{Test statistic } T = \frac{180 - 175}{55.042 \sqrt{\frac{1}{15} + \frac{1}{14}}} = 0.24$$

Since the test statistic lies in the critical value region, H_0 is not rejected. Therefore, the average expenditure on home insurance is the same in both regions.

Confidence Interval and Hypothesis Testing for a Single Proportion.

Confidence Interval for a Single Proportion.

It is of interest to estimate the proportion of employees, who favour a certain type of work, or the proportion of defective items in a certain lot, or the proportion of distance students who are in formal employment. Let P be the true proportion of elements that have attribute A (a

certain characteristic of interest) in a population. We draw a simple random sample of size n from this population and Let X be the number of elements in the sample that have attribute A .

Thus the point estimate of the true proportion p in the population is given by $\hat{p} = \frac{X}{n}$, where

X has a binomial distribution with parameters n and p . Recall that $E(X) = np$, and $\text{Var} = npq$.

Hence we find that $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = \frac{pq}{n}$. By the Central Limit Theorem, the

random variable given $Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$ has a standard normal distribution as n increases.

Therefore the $100(1 - \alpha)$ confidence interval for the population proportion P is given by

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \text{ where } Z_{\frac{\alpha}{2}} \text{ is the critical value from the normal distribution}$$

Example 6.25

In a simple random sample of 500 students, 160 preferred (to) taking classes in the morning rather than in the afternoon. Construct a 95% confidence interval for the true proportion of employees who favour morning training classes.

Solution

$$\hat{p} = \frac{160}{500} = 0.32 \text{ therefore } \hat{q} = 1 - 0.32 = 0.68 \text{ critical value} = 1.96$$

$$95\% \text{ confidence interval} = 0.32 \pm 1.96 \sqrt{\frac{0.32 \times 0.68}{500}}$$

$$= 0.32 \pm 0.04$$

$$= (0.32 - 0.04) \text{ to } (0.32 + 0.04)$$

$$= 0.28 \text{ to } 0.36$$

This means that the true proportion, of the students who favour the morning classes, is between 28% and 36%

Hypothesis Testing for a Single Proportion

We have seen how to conduct hypothesis tests for a mean. We now turn to proportions. The process is completely analogous, although we will need to use the standard error formula for a proportion.

We may wish to test whether a random sample of size n , with proportion of 'successes' p_s could have been drawn from a population with proportion of successes p .

We use the test statistic given by

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

where the null hypothesis is that the proportion of successes in the population is p .

The hypothesis can take any of the form;

$$\begin{array}{lll} H_0 : P = a\% & H_0 : P \geq a\% & H_0 : P \leq a\% \\ H_1 : P \neq a\% & \text{or } H_1 : P < a\% & \text{or } H_1 : P > a\% \end{array}$$

Example 6.26

An auditor claims that 10% of invoices for a certain company are incorrect. To test this claim, a random sample of 200 invoices is checked and 24 are found to be incorrect. Test at the 1% significant level to see if the auditor's claim is supported by the sample evidence.

Solution

$$H_0 : P = 0.1$$

$$H_1 : P \neq 0.1$$

The critical value range is from -2.58 to 2.58 since it is a two tailed test.

$$\hat{p} = \frac{24}{200} = 0.12$$

$$\text{Test statistic } Z = \frac{0.12 - 0.1}{\sqrt{\frac{0.1 \times 0.9}{200}}} = 0.94$$

Since the test statistic lies in the critical value region, H_0 is not rejected, therefore the auditors' claim is supported.

Example 6.27

A bank manager wishes to test the hypothesis that 50% or more of the bank's loans are made to customers who reside in the town where the bank is located. A researcher selects a random sample of 250 of the bank's borrowers and finds that 43% of them reside in the town. If a 5% significance level is used, should the bank manager's claim be rejected?

$$H_0 : P \geq 0.5$$

$$H_1 : P < 0.5$$

$$\hat{p} = 0.43 \quad n = 250$$

$$\text{Test statistic } Z = \frac{0.43 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{250}}} = -2.21$$

Since the test statistic is less than the critical value, H_0 is rejected, therefore the bank managers' claim can be rejected.

Confidence Interval and Hypothesis Testing for a Difference Between Two Proportions

Confidence Interval for a Difference Between Two Proportions

Comparisons of proportions, in different groups, are a common practice. A wholesaler compares the proportions of defective items found in two separate sources of supply from which he buys these items. A safety engineer compares the proportions of head injuries sustained in a car accident by passengers with seat belts against those without seat belts.

Consider two independent samples of sizes n_1 and n_2 that are drawn from two binomial populations with parameters p_1 and p_2 . A $100(1 - \alpha)\%$ confidence interval will be derived on the difference between p_1 and p_2 using the central limit theorem.

Let x_1 and x_2 be the number of successes obtained in sample 1 and sample 2 respectively.

We then have $\hat{P}_1 = \frac{x_1}{n_1}$ and $\hat{P}_2 = \frac{x_2}{n_2}$ as the point estimates of p_1 and p_2 respectively.

Moreover we have:

$$E(\hat{P}_1) = p_1, E(\hat{P}_2) = p_2, \text{ with } Var(\hat{P}_1) = \frac{p_1 q_1}{n_1} \text{ and } Var(\hat{P}_2) = \frac{p_2 q_2}{n_2}$$

Because of the independence of the two samples, we can write

$$E(\hat{P}_1 - \hat{P}_2) = p_1 - p_2 \text{ with } Var(\hat{P}_1 - \hat{P}_2) = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$$

Now by Central Limit Theorem, the random variable Z is given by

$$Z = \frac{(\hat{P}_1 - \hat{P}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}} \text{ has a standard normal distribution hence the } 100(1 - \alpha)\% \text{ confidence interval for the difference between the two proportions, } p_1 \text{ and } p_2 \text{ is given by;}$$

$$(\hat{P}_1 - \hat{P}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

Example 6.28

A certain change in a manufacturing procedure for component parts is being considered. Samples were taken using both the old and new procedures in order to determine the difference induced by the new procedure. Suppose that 75 out of 1500 items, from the existing procedure, were found defective, and 80 out of 2000 items, from the new procedure were found to be defective. Find a 98% confidence interval for the true difference in the fraction of defectives between the two procedures.

Solution

$$\hat{p}_1 = \frac{75}{1500} = 0.05 \quad \hat{p}_2 = \frac{80}{2000} = 0.04$$

$$95\% \text{ confidence interval} = (0.05 - 0.04) \pm 2.33 \sqrt{\frac{0.05 \times 0.95}{1500} + \frac{0.04 \times 0.96}{2000}}$$

$$= 0.01 \pm 0.02$$

$$= (0.01 - 0.02) \text{ to } (0.01 + 0.02)$$

$$= -0.01 \text{ to } 0.03$$

Hypothesis Testing for a Difference Between Two Proportions

If two samples are counted independently of each other, we use the test statistic given below, i.e we test whether the samples have been drawn from populations which have a common proportion p .

$$Z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \text{ when } p \text{ the population proportion is known and}$$

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \text{ where } \hat{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \text{ when } p \text{ is unknown where } \hat{p} \text{ is the estimator of it.}$$

There are three ways to set up the null and alternative hypotheses.

- Equal hypothesis versus not equal hypothesis: $H_0 : P_1 = P_2$ versus $H_0 : P_1 \neq P_2$, two-tailed test
- At least versus less than: $H_0 : P_1 \geq P_2$ versus $H_0 : P_1 < P_2$, left-tailed one-tailed test
- At most versus greater than: $H_0 : P_1 \leq P_2$ versus $H_0 : P_1 > P_2$, right-tailed one-tailed test

Example 6.29

Distance Students from region A claim that their percentage of passed finalists is always higher than that of region B. Two samples of final year candidates who sat at five examination centres were randomly selected from each region for the previous examinations and the statistics are as follows;

	Region	
	A	B
Number of Candidates	80	60
Number of passed finalists	64	45

Test the claim made by region A candidates at 2% level of significance.

Solution

$$H_0 : P_1 = P_2$$

$$H_1 : P_1 > P_2$$

$$P_1 = \frac{64}{80} = 0.8 \qquad P_2 = \frac{45}{60} = 0.75$$

$$\hat{p} = \frac{80 \times 0.8 + 60 \times 0.75}{80 + 60} = 0.78 \qquad \text{therefore } \hat{q} = 0.22$$

$$\text{Test statistic } Z = \frac{0.8 - 0.75}{\sqrt{0.78 \times 0.22 \left(\frac{1}{80} + \frac{1}{60} \right)}} = 0.71$$

Since the test statistic (0.71) is less than the critical value(2.05), H_0 is not rejected.

Therefore the claim by region A candidates is justifiable.

6.9.7 Regression Analysis

An essential aspect of managing any organization is planning for the future. This leads to budgeting, whose aim is to meet future financial and non-financial information. The long-run success of an organization is closely related to how well management is able to anticipate the future and develop appropriate strategies. Good judgment, intuition and an awareness of the state of the economy may give a manager a rough idea of what is likely to happen in the future. Converting this into a number can be used as next quarter's sales volume or next year's raw material quote per unit, which is often difficult. This is the purpose of *forecasting*.

A forecast is a prediction of what will happen in the future. There are several forecasting techniques but we will only look at the scatter diagram method, linear regression analysis, and time series analysis.

Linear regression analysis is a technique used to show the relationship between an independent (units/volume) variable and a dependent variable (costs). This relationship is in the form of an equation of a straight line $y = a + bx$, where;

y- dependent variable

x - independent variable

a - y intercept (the line cuts the y axis)

b - gradient

Given that the linear regression equation is $y = a + bx$

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \text{ and } a = \frac{\sum y}{n} - \frac{b \sum x}{n} \text{ where } n \text{ is the number of } (x, y) \text{ variables.}$$

Once the values of a and b have been found, the regression equation can then be used to forecast y, given a value of x for any period, hence future costs (y) can be estimated given a level of activity (x) and variable costs per activity (b) and fixed cost (a).

Example 6.30

It is believed that a relationship exists between the price of a house and its distance from the city centre. A sample of seven houses was collected and the distances from the town centre and their prices were as follows:

Distance (kilometres)	Price (K'000)
1	72
2	70
3	69
5	68
4	45
6	38
7	40

a) Calculate the least squares regression equation for the two variables.

- b) Using your result from (a) above, estimate the price for a house located 8.5 kilometres away from the city centre.

Solution

a)

x	y	xy	x^2
1	72	72	1
2	70	140	4
3	69	207	9
5	68	340	25
4	45	180	16
6	38	228	36
7	40	280	49
$\sum x = 28$	$\sum y = 402$	$\sum xy = 1447$	$\sum x^2 = 140$

b =

$$\frac{(7 \times 1447) - (28 \times 402)}{(7 \times 140) - 28^2} = \frac{-1127}{196} = -5.75$$

$$a = \frac{402}{7} - (-5.75) \frac{28}{7} = 80.43$$

Therefore the regression equation is $y = 80.43 - 5.75x$

b) $y = 80.43 - 5.75(8.5) = 31.555$ i.e., K31555 or K32000

6.9.8 Correlation

Correlation measures the strength of the relationship between two variables. Positive correlation is when the increase in one variable results in an increase in the other variable and vice-versa. Negative correlation is when an increase in one variable results in a decrease in the other variable and vice versa. This is measured using the coefficient of correlation of which there are two types.

Pearson's Product Moment Correlation Coefficient

This is a measure of linear correlation.

It is given by

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \times \sqrt{n \sum y^2 - (\sum y)^2}}$$

Here are some properties of r

r only measures the strength of a linear relationship. There are other kinds of relationships besides linear.

r is always between -1 and 1 inclusive. -1 means perfect negative linear correlation and +1 means perfect positive linear correlation

r has the same sign as the slope of the regression (best fit) line

r does not change if the independent (x) and dependent (y) variables are interchanged

r does not change if the scale on either variable is changed. You may multiply, divide, add, or subtract a value to/from all the x-values or y-values without changing the value of r.

The closer r is to 1, the stronger is the evidence of positive association between the two variables.

The closer r is to -1, the stronger is the evidence of negative association between the two variables.

The closer r is to 0, the little or no evidence of a linear relation between the two variables.

Example 6.31

As an example, we shall consider whether there is any relationship between 'Ice cream Sales' for an ice-cream manufacturer and 'Average Monthly Temperature'. The collected data is recorded below:

<u>Month</u>	<u>Av.Temp.</u>	<u>Sales</u>
	(°C)	(K'000)
January	4	73

February	4	57
March	7	81
April	8	94
May	12	110
June	15	124
July	16	134
August	17	139
September	14	124
October	11	103
November	7	81
December	5	80

Solution

x	y	xy	x^2	y^2
4	73	292	16	5329
4	57	228	16	3249
7	81	567	49	6561
8	94	752	64	8836
12	110	1320	144	12100
15	124	1860	225	15376
16	134	2144	256	17956
17	139	2363	289	19321

14	124	1736	196	15376
11	103	1133	121	10609
7	81	567	49	6561
5	80	400	25	6400
120	1200	13362	1450	127674

$$r = \frac{12 \times 13362 - 120 \times 1200}{\sqrt{(12 \times 1450 - 120^2) \times (12 \times 127674 - 1200^2)}} = 0.98$$

Therefore, there is a strong positive correlation between average temperature and sales.

Spearman's Rank Correlation Coefficient

It is used when the values of the variables are only ranks, or positions.

For each of n pairs of ranks, compute the difference in ranks, d, ignoring minus signs and it is given by

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Example 6.32

Calculate Spearman's correlation coefficient for the variables in example 6.31

	x	y	R_x	R_y	d	d^2
1	72	1		7	-6	36
2	70	2		6	-4	16
3	69	3		5	-2	4
5	68	5		4	1	1
4	45	4		3	1	1
6	38	6		1	5	25
7	40	7		2	5	25

$$\sum d^2 = 108$$

$$\sum d^2 = 108$$

$$r_s = 1 - \frac{6 \times 108}{7(7^2 - 1)} = 1 - \frac{648}{336} = -0.93$$

There is therefore strong negative correlation between sales of ice-cream and temperature changes.

6.9.9 Type I and Type II errors

Saunders *et al.* (2009, p.452) assert that there are two types of errors that can result when you make inferences from samples. **Type I error** occurs when a reliable theory is incorrectly falsified through unreliable observation and testing i.e. wrongly concluding that something is true when in fact not. This type of error tends to occur in the early stages of research into a new field. For example, when a new disease such as HIVAIDS appears, researchers do not know much about the disease, and they may establish experiments that appear reliable but in fact are not. Because of incorrect assumptions, the experimentation may be measuring the wrong variable. As a result, a theory linking HIV-AIDS to contaminated intravenous injections might be incorrectly falsified.

Conversely, a Type II error occurs when an unreliable theory is not falsified because reliable observations are considered to be unreliable i.e. wrongly concluding that something is not true when it is true. In this case, a false theory is in fact not falsified

because reliable falsification results are disregarded. This error often occurs when somebody first develops an important linkage between two variables.

6.9.10 Required reading 6.2



Now read Chapter 12 in Saunders *et al.* (2009).

6.9.11 Activity 6.2



Attempt the self-check and discussion questions in Chapter 12 in Saunders *et al.* (2009).

6.10 DATA PROCESSING AND ANALYSIS SOFTWARE

There are many software packages that you can use to process and analyse qualitative and quantitative data. The common ones are Atlas.ti and Nvivo for qualitative data and SPSS for quantitative data. The software requires training which is beyond the scope of this course. However, you can use spreadsheets such as Microsoft Excel to carry out basic data processing and analysis of your qualitative data (e.g. content analysis) and quantitative data (e.g. drawing charts and graphs).

6.11 SUMMARY



This unit has introduced you to the techniques that you need to process, present and analyse your data to enable you to answer your research questions. The techniques that you choose depend on whether your data is qualitative or quantitative. Either way you can use simple techniques such as spread-sheets or data analysis software to analyse your data.

For quantitative data, descriptive statistics such as frequency distributions, charts, measures of central tendency and measures of dispersion help you to process and present your data. Inferential statistics such as distributions, confidence intervals, regression analysis, and correlation analysis help you to analyse your data.

With respect to qualitative data, thematic analysis can be used to identify themes that are used to categorise the data for further analysis e.g. by using content analysis. Other analysis techniques such as grounded theory or analytic induction involve an iterative process between theory generation and data collection, and may be beyond the resources available to an undergraduate researcher.

7.0 UNIT SIX: WRITING THE DISSERTATION

7.1 INTRODUCTION



Even though this unit on writing and presenting your project report is the last one in this module, the process of writing goes on throughout the project. Writing helps you to think and clarify your thoughts. It is easier to say something than to write it down! This unit helps you to write your dissertation by explaining the contents of the dissertation and what you are expected to write in each section. These include cover page, abstract/executive summary, contents page, introductory chapter, literature review, methods section, data analysis/findings, discussion, conclusions and recommendations, references and bibliography and appendices. There is also advice on the writing style that you should adopt to improve the readability of your project report. You should adopt a straightforward, formal style, avoiding figures of speech, idioms, among others.

7.2 AIM



The aims of this unit are to introduce you to the standard contents of a dissertation and advise you on issues you need to consider when writing and presenting your dissertation.

7.3 OBJECTIVES



By the end of this unit, and having completed the prescribed readings and activities, you should be able to:

Explain the dissertation structure including cover page, abstract/executive summary, contents page, introductory chapter, literature review, methods section, data analysis/findings, discussion, conclusions and recommendations, references and bibliography, and appendices.

7.4 TIME REQUIRED



You should spend a minimum of 5 hours on this unit. This time should be spent on reading the contents of this unit, reading the prescribed reading [Chapter 14 in Saunders *et al.* (2009) and/or Chapter 27 in Bryman and Bell (2011)] and on attempting self-check and discussion questions in the prescribed readings itemized above.

7.5 REFLECTION



You have decided what your research is about. You have reviewed the literature and clarified your research objectives and research questions. You have collected, presented and analysed the data you need so that you can answer your research questions. Now you should write your project report. What is your rhetorical strategy? What writing style should you adopt so that your report is persuasive and convincing?

7.6 DISSERTATION STRUCTURE

The structure of your dissertation may vary slightly depending on whether your research approach is inductive or deductive (Saunders *et al.* (2009, p.531). However, I highly recommend that you adopt the sections explained below in the order in which they appear as it is the general structure adopted for business and management research dissertations.

7.6.1 Cover/title page

This should have your university logo and name, dissertation title, your name, your student number, title of degree, the name of your supervisor, the deadline date, and word count.

7.6.2 Acknowledgements

It is usual but not compulsory to thank those who have been of particular help to you in completing the dissertation.

7.6.3 Abstract

This is a short section which concisely summarises the whole of the dissertation. The key points that you should include are the aim of the research, research objectives and research questions, the research design, the findings of your research and the conclusions you have drawn. You should write the abstract in the past tense. The word count should not exceed 300 words.

7.6.4 Introduction Chapter

Your introduction should contain your topic clearly stated and defined, the reason why it is of interest to you, a clear overall purpose and definitions of all special and general terms. It should also include background history of the organization chosen, a clear research question or questions, a list of objectives, and a hypothesis or hypotheses if needed.

7.6.5 Literature Review Chapter

This chapter should demonstrate that you have conducted a thorough and critical investigation of relevant sources, outlining, comparing and discussing key ideas, explanations, concepts, models and theories. You should present these ideas in a systematic, well-structured and logical sequence. You will be expected to use prominent and up-to-date books and academic journals. This chapter should end with a statement of the gap in current knowledge which your research aims to fill. You will need to demonstrate here the scope of your reading and authorities cited from a variety of sources.

7.6.6 Research Methodology Chapter

This chapter describes and assesses the approach you have taken to the data collection process (research approach, research strategy, method(s), validity etc.) For each research objective and question or hypothesis, you should have a method for achieving it, making sure that you offer clear rationale for the decisions that you have made. You should explicitly describe your chosen method(s) and any sampling techniques used. It is also important to give a brief assessment of other potential relevant data collection methods and

why you discounted them. Do not describe all data collection methods. There should be a critique of the success, or otherwise, of your method(s). Explain the appropriateness of the data analysis techniques that you have selected. You must also discuss validity, reliability and generalisability.

7.6.7 Findings and Analysis Chapter

In this chapter the data generated should be reported as completely and neutrally as is possible such that the reader can assess it easily. This is where you will include such tables and graphs that will illustrate your findings. This chapter will also contain verbatim quotes from interviewees, or sections of narrative account that illustrate periods of unstructured observation.

You should present your analysis clearly and logically and it should be relevant to your research aim, research question(s), hypothesis(es) and objectives. Make sure that you relate the findings of your primary research to your literature review. You can do this by comparison: discussing similarities and particularly differences. If you think your findings have confirmed some literature findings, say so and say why. If you think your findings are at variance with the literature, say so and say why. Ensure that tables, charts, graphs, and other illustrations are clearly labeled as Figure 1, or Table 1 etc., and indicate their sources.

7.6.8 Conclusions and Recommendations Chapter

State the main conclusions of your dissertation. State explicitly how and to what extent you have met your aims and objectives / answered your research question(s) / proved your hypothesis(es), whichever is appropriate. Your conclusions should follow logically from your findings and not contain any new material. Recommendations in bullet point form can be made if appropriate.

7.6.9 References and Appendices

Give full references to every source used, presented in the format of the Harvard System of Referencing. Any necessary information should be here, for example, sampled questionnaires, topic guides, etc. Each appendix should be numbered and should consist of detailed information that is interesting but not essential to the main thrust of your findings section.

7.7 WRITING STYLE

7.7.1 Clarity and simplicity

Saunders *et al.* (2009, pp.545-547) give the following advice on how to effectively communicate with the readers of your project report.

- ✓ Avoid very long sentences; instead write short, simple sentences.
- ✓ Explain your technical terms in a glossary in one of the appendices and avoid unnecessary jargon.
- ✓ Use direct quotes sparingly; instead paraphrase cited material to demonstrate understanding.
- ✓ Proof-read your work to correct spelling, grammar and punctuation errors before submitting it for marking.

7.7.2 Person, tense and gender

Academic writing has traditionally been impersonal, in the past tense and uses the passive voice (Saunders *et al.*, 2009, p.547). You are encouraged to use the past tense when writing up your project report. You should avoid writing in the first person (do not use 'I' 'my' etc.); instead write in the third person (e.g. 'the research...' instead of 'my research...'). You should also be gender sensitive and avoid portraying one sex as being more superior to another e.g. by assuming that people with certain positions or titles are male or female.

7.7.3 Preserving anonymity of research participants

One of the ethical issues that normally arises in research is confidentiality of information. To deal with this threat, you may have undertaken to conceal the identity of the organisation and the research participants. Saunders *et al.* (2009, p.548) advise that you can achieve this by using fictitious names for the organisations and by avoiding referring to individual participants by name.

7.7.4 Continual revision of the project report

Many students do not seem to distinguish between an examination and dissertation in terms of the quality of work expected of them. An examination is often written under extreme time pressure and the script is probably read by one tutor and not retained for use by other

students. The converse is true regarding your dissertation, which is likely to be read by many more people. Saunders *et al.* (2009, p.548) therefore advise that you should read your dissertation and polish it up several times before making the final submission.

7.8 REQUIRED READING 7.1



Now read Chapter 14 in Saunders *et al.* (2009) and/or Chapter 27 in Bryman and Bell (2011).

7.9 ACTIVITIES



Attempt the self-check and discussion questions in Chapter 14 in Saunders *et al.* (2009).

7.10 SUMMARY



This unit has introduced you to the contents of your dissertation and what you are expected to write in each section. These include cover page, abstract/executive summary, contents page, introductory chapter, literature review, methods section, data analysis/findings, discussion, conclusions and recommendations, references and bibliography, and appendices.

This unit has further given you advice on the writing style that you should adopt to improve the readability of your project report. The advice is in the areas of clarity and simplicity in your writing, and the need for you to write in past tense, use third person style and avoid sexist language. You have also been advised to preserve the identity and confidentiality of your research participants, and continually revise your draft project reports to correct grammar, spelling, and punctuation errors. Ensure that pages, sections and chapters are correctly numbered.

7.11 RESEARCH PROPOSAL SUBMISSION



You should now be finalizing your research proposal in readiness for submission. Your research proposal is due during the third week of June (if you are taking this course during the January to June semester) or the third week of December (if you are taking this course during the July to December semester). Verify the actual submission deadline on Moodle or the academic calendar or contact your course lecturer.

The submission deadline for your research proposal falls after the residential school. You should therefore take advantage of this opportunity to meet with your research supervisor and discuss and agree your research topic, research objectives and research questions. This is very important because you cannot easily change these fundamental aspects of your research next semester.

Note that your research proposal accounts for 50% of your total marks for this course. The word count should be 2,000 words +/- 10%. Failure to meet this word count attracts penalties. You should use the Times New Roman font, size 12, line spacing of 1.5 and justify the text. You should submit your assignment through TurnItIn. Do not e-mail the assignment to the lecturer.

LIST OF REFERENCES

1. Adams, J. Khan, H. T. A., Raeside, R. and White, D. (2007) *Research Methods for Graduate Business and Social Science Students*. London: Sage.
2. Bryman, A. and Bell, E. (2011) *Business Research Methods*, 3rd Edition. New York: Oxford University Press.
3. Cohen, L., Manion, L and Morrison, K. (2007) *Research Methods in Education*, 6th edition. London: Routledge Ltd.
4. Emerald Group Publishing (2014) Advanced Search [Online]. Retrieved from: <http://www.emeraldinsight.com/search/advanced>. Accessed on 16 December 2016.
5. Gall, M. D., Gall, J. P. and Borg, W. (2006) *Educational Research: An Introduction* (8th Edn). New York: Longman. Cited in Saunders, N. K. M, Thornhill, A. and Lewis, P. (2009) *Research Methods for Business Students*, 5th Edition, London: Apprentice Hall.
6. Google Scholar (2014) *Google Scholar* [Online]. Retrieved from: <http://scholar.google.com/>. Accessed on 16 December, 2014.
7. Ryan, G. W. and Bernard, H. R. (2003) Techniques to Identify Themes, *Field Methods*, Vol. 15, No. 1, pp. 85–109.
8. Robson, C. (2002) *Real World Research*, 2nd Edition. Oxford: Blackwell.

9. Saunders, N. K. M, Thornhill, A. and Lewis, P. (2009) *Research Methods for Business Students*, 5th Edition, London: Apprentice Hall.
10. Walliman, N. (2005) *Your Research Project: A step by Step Guide for the First-Time Researcher*, 2nd Edition. London: Sage.
11. Yin, R. K. (1981) 'The Case Study Crisis: Some Answers'. *Administrative Science Quarterly*, Volume 26.

APPENDIX 1 BFS 421 RESEARCH METHODS COURSE OUTLINE

Pre-requisite: None

AIM

The aim of this course is to enable students understand and use quantitative and qualitative research techniques to answer a research question, as appropriate to final-year undergraduate level.

OBJECTIVES

By the end of the course, students should be able to:

6. Understand and explain the theoretical underpinning and practical implications of research conducted at final-year undergraduate level
7. Utilise qualitative and quantitative research methods to answer a research question.
8. Compare alternative research approaches and evaluate their uses and limitations.
9. Understand and discuss the key concepts and tasks involved in a research project and how these interrelate in a complementary way to form a coherent research proposal.
10. Process and analyse research data using data analysis software.

CONTENT

PART A THE ROLE OF RESEARCH AND ETHICS

1. Introduction to research
2. Nature and purpose of research, including business research
3. Overview of the research process
4. Choosing a research topic
5. Role of ethics in business research e.g. confidentiality, plagiarism and cheating

PART B RESEARCH PROPOSAL

1. Content of proposal e.g. topic, background/context, research question/aim/objectives, literature review, methodology, role of ethics, timescale and resources.
2. The research topic including attributes of a good research topic, generating and refining research ideas.
3. Research aim, objectives and questions.

PART C LITERATURE REVIEW

1. Purpose of a literature review
2. Literature sources
3. Literature search, including defining parameters and generating key words/terms, conducting the literature search, and use of databases in libraries
4. Evaluating the literature and assessing relevance and sufficiency
5. Critically reviewing the literature
6. Citations, bibliography, and references including use of Harvard and APA systems of referencing.

PART D RESEARCH METHODOLOGY

1. Types of research e.g. descriptive, explorative and explanatory studies.
2. Research methods - quantitative versus qualitative research.
3. Different approaches to research - inductive versus deductive research.
4. Different research strategies including experimental, survey, case study, grounded theory, action research, cross-sectional, exploratory, explanatory and descriptive studies.
5. Data collection tools or instruments including sampling, observation, interviews, questionnaires, documents, and secondary data.

PART E DATA ANALYSIS

1. Qualitative data analysis including thematic and content analysis.
2. Reliability, validity, and generalizability of qualitative results.
3. Quantitative data presentation e.g. presentation of graphs and charts.
4. Descriptive statistics e.g. distribution, dispersion and central tendency of the variable.
5. Causation versus correlation.
6. Inferential statistics including normal and non-normal distributions, hypothesis testing, confidence intervals, Type I and II errors.
7. Validity and precision of quantitative results.

8. Testing for significant relationships.
9. Use of common data processing and analysis software such as Stata, Matlab, R, Eviews, Nvivo, Atlas.ti and SPSS.

PART F WRITING THE DISSERTATION

Dissertation structure, including cover page, abstract/executive summary, contents page, introductory chapter, literature review, methods section or methodology, data analysis/findings, discussion, conclusions and recommendations, references and bibliography, and appendices.

METHOD OF TEACHING

Three hours of lectures and one hour tutorial per week.

ASSESSMENT

Continuous assessment	50%
2 assignments of equal weight	50%
Research Proposal	<u>50%</u>
Total	<u>100%</u>

The Research Proposal should have 2,000 words +/- 10%, Times New Roman font size 12, single spacing with justified text.

PRESCRIBED READING

1. Bryman, A. and Bell, E. (2011) *Business Research Methods*, 3rd Edition. New York: Oxford University Press.
2. Patton, R. (2001) *Qualitative Evaluation & Research Methods*. Sage.
3. Saunders, N. K. M, Thornhill, A. and Lewis, P. (2009) *Research Methods for Business Students*, 5th Edition, London: Apprentice Hall.

RECOMMENDED READING

9. Allison, B. (2004) *The student's guide to preparing dissertations and theses*. Routledge.
10. Bell, J. (2008) *Doing your research project: a guide for first time researchers in education and social science*. Open University Press.

11. Collins, J. & Hussey, R. (2003) *Business Research: A practical guide for undergraduates and Postgraduates students*. Palgrave
12. Fisher, C. (2004) *Researching and writing a dissertation for business students*. FT Prentice Hall
13. Gash, S. (2000) *Effective literature searching for research*. Gower
14. Hart, C. (2001) *Doing a literature Search*. Sage.
15. Kumar, R. (1999) *Research methodology: a step-by-step guide for beginner*. Sage.

APPENDIX 2 EXAMPLES OF TOPIC AREAS AND TITLES (adapted from OBU, 2014 and AIT, 2008)

1. A case study of the development of professional judgement in the area of off-balance sheet liabilities.
2. A case study on accountability in a publicly-funded body.
3. A case study to assess the economic impact on the Zambian economy of specified tax exempt activities, e.g. film production, stallion and greyhound services and the arts.
4. A critical analysis of the historical development of the conceptual accounting framework.
5. A critical assessment of the financial and non-financial effects of the public sector benchmarking process.
6. A critical evaluation of the financial and operational risk management within an organisation.
7. A critical evaluation of the planning and implementation of an information system in an organisation.
8. A critical evaluation of the restructuring of an organisation's operational activities and the effect on the organisation's financial performance.
9. A critical review of key factors or indicators in the motivation of employees in an organisation.
10. A critique of intellectual property valuation methods.

11. A review of the business management and accounting issues of environmental costs of an organisation.
12. A review of the effectiveness of the use of costing techniques within an organisation.
13. A review of the marketing strategy of an organisation and its effectiveness.
14. A synthesis of client criteria in the assessment of choice of professional firm.
15. A synthesis of the impact of sourcing alternative public funding for specific projects (other than from direct or indirect taxation mechanisms)
16. Accounting policy choice in the area of financial instruments and derivatives.
17. Accounting, control and culture: a social analysis of change in three indigenous Zambian companies.
18. An analysis and evaluation of an organisation's budgetary control system and its links with performance management and decision making.
19. An analysis and evaluation of the business and financial performance of an organisation over a three year period.
20. An analysis and evaluation of the financial and operational consequences of a merger between two organisations or of the acquisition of one organisation by another.
21. An analysis and evaluation of the management of an organisation's working capital over a three year period and its impact on the organisation's funding strategies.
22. An appraisal of the business and financial objectives of a strategic investment decision made by an organisation and its impact on key stakeholders.
23. An assessment of an organisation's corporate social responsibility policies, including business ethics, and their impact on business practice and key stakeholders.
24. An assessment of the costs and benefits of the emerging development of shared financial services centres.
25. An assessment of the impact of executive information systems on management

decision making.

26. An assessment of the potential impact of an aspect of impending legislation on the operations and financial position of an organisation.
27. An assessment of the quality of the corporate governance within an organisation and the impact on an organisation's key stakeholders.
28. An evaluation of how the introduction of a new technology can assist an organisation in achieving its business objectives.
29. An evaluation of the contribution made by human resource activities to the attainment of business and financial objectives.
30. An evaluation of the perception of professional skill development in the wider accounting profession.
31. An evaluation of the use of short term and long term Islamic financial instruments and their impact on the financial statements of an organisation.
32. An investigation into the financial and operational costs and benefits of the internal audit / internal review activities within an organisation.
33. An investigation into the possible effects of a proposed accounting standard on the financial statements and business activities of an organisation.
34. Developing public sector accountable management systems – a case study of Zambia's IFMIS.
35. Economic determinants of accounting policy choice and asset management.
36. Ethics and discretion in the provision of public finance estimates.
37. Influences on the selection and disclosure of accounting policies in a selected developing country.
38. Risk management and safety improvement within the airline industry.
39. SERVQUAL model and factors affecting customer satisfaction in Zambian banking.

40. Three laws regarding the operation of brands – theories, foundational and case analysis on brand operation.
41. Understanding consumers' attitudes towards loyalty and loyalty cards in Zambia.

References

1. Oxford Brookes University (2014) *BSc (Hons) in Applied Accounting and Research and Analysis Project Information Pack*. Oxford: Oxford Brookes University.
2. Athlone Institute of Technology (2009) *Bachelor of Business in Applied Accounting Course Manual*. Athlone: Athlone Institute of Technology.